

SUBMODULAR MAXIMIZATION WITH PREDICTIONS: EXACT WORST CASES, QUERY HARDNESS, AND PER- RUN CERTIFICATES

Anonymous

ABSTRACT

1 INTRODUCTION

2 MODEL AND PRELIMINARIES

A ground set N with $|N| = n$, a monotone submodular $f : 2^N \rightarrow \mathbb{R}_{\geq 0}$ with $f(\emptyset) = 0$, and a cardinality budget K are fixed. The algorithm cannot evaluate f ; it can only query a predictor $\tilde{f} : 2^N \rightarrow \mathbb{R}$ with $\tilde{f}(\emptyset) = 0$. Marginal gains are written $d_e(S) = f(S \cup \{e\}) - f(S)$ and $\tilde{d}_e(S) = \tilde{f}(S \cup \{e\}) - \tilde{f}(S)$.

Definition 1 (Prediction error). *For $\eta_u, \eta_o \geq 1$ the predictor $\tilde{f}$ has (single-element, multiplicative) error at most (η_u, η_o) if for every $S \subseteq N$ and $e \notin S$,*

$$d_e(S)/\eta_u \leq \tilde{d}_e(S) \leq \eta_o d_e(S).$$

In particular $d_e(S) = 0$ forces $\tilde{d}_e(S) = 0$, and all predicted gains are nonnegative. The scalar error is $\eta = \eta_u \eta_o$.

Approximation ratios are stated as $\alpha \in (0, 1]$ with $F^{\text{ALG}} \geq \alpha F^{\text{OPT}}$; larger is better.

Predictive greedy is the single-step greedy run on $\tilde{f}$: for $t = 0, \dots, K - 1$ it adds an element maximizing $\tilde{d}_e(S^t)$, with ties broken adversarially in all worst-case statements. Its exact worst-case ratio at error level η is written $\rho_K(\eta)$. Two run-dependent error measures refine η :

Definition 2 (Selection error). *For an execution of predictive greedy with states $S^0, \dots, S^{K-1}$ and picks $e_0, \dots, e_{K-1}$, the selection error is*

$$\eta^{\text{sel}} = \max \left\{ \max_{e \notin S^t} d_e(S^t) / d_{e_t}(S^t) : t < K, d_{e_t}(S^t) > 0 \right\} \geq 1,$$

the largest factor by which a step of the run fell short of the best true gain, over the steps with positive chosen gain. In the terminology of Goundan and Schulz, a run with selection error η^{sel} is an execution of greedy with an α -approximate incremental oracle with $\alpha = \eta^{\text{sel}}$ (Goundan & Schulz, 2007).

The trajectory error restricts Definition 1 to the states of the run; it is used in a remark and in the appendix, and written η^{tr} when a symbol is needed. The chain $\eta^{\text{sel}} \leq \eta^{\text{tr}} \leq \eta$ holds for every run.

Table 2 in Appendix A collects the notation.

3 RELATED WORK

Approximate incremental oracles. The closest error model replaces the exact greedy step by an oracle returning an element whose true marginal gain is within a factor of the best available one, the α -approximate incremental oracle of Goundan & Schulz (2007), whose $\alpha \geq 1$ is exactly the selection error η^{sel} of Definition 2. Their Theorem 1 gives $1 - (1 - 1/(\alpha k))^k \geq 1 - e^{-1/\alpha}$ under

such an oracle, which at $\alpha = 1$ is the classical cardinality-constrained bound of Nemhauser et al. (1978), unimprovable by polynomially many value queries (Nemhauser & Wolsey, 1978); for the related set cover problem no polynomial-time algorithm approximates within $(1 - \epsilon) \ln n$ unless $\text{NP} \subseteq \text{DTIME}(n^{O(\log \log n)})$ (Feige, 1998). We differ in the question, not the bound: Proposition 4 is their theorem restated in the model of Section 2 and attributed as such, while what is new is the exact worst-case ratio at every error level (Theorem 8), a per- K attaining family at error η^{sel} (Theorem 6), and a matching bounded-query obstruction (Theorem 12).

Approximate submodularity. A second line weakens the objective instead of the access, measuring distance from submodularity by the submodularity ratio γ of Das & Kempe (2011, Definition 2.3). Greedy then still returns a $(1 - e^{-\gamma})$ fraction of the optimum (Das & Kempe, 2011, Theorem 3.2), and Elenberg et al. (2018) derive weak submodularity from restricted strong convexity, extending constant-factor greedy guarantees beyond submodular objectives. There the oracle is exact and the structure approximate, here the structure is assumed and the access is not; the third experiment family of Section 5 lies in the intersection, its true objective being measurably neither monotone nor submodular.

Noise and inexact oracles. A third line perturbs the value oracle itself, by random noise or by allowing the queried function to be only approximately submodular. Hassidim & Singer (2017) recover a ratio arbitrarily close to $1 - 1/e$ under independent noise once the budget is large enough, and state that adversarial noise admits no non-trivial guarantee; Horel & Singer (2016, Theorem 3) show that for ϵ -approximately submodular functions with ϵ above roughly $n^{-1/2}$ no algorithm with polynomially many queries beats $O(n^{-\delta})$; and Bhawalkar et al. (2025) treat a persistent noisy value oracle, where repeated queries to a set return one value, obtaining $1 - 1/e$ for monotone objectives under a matroid constraint, $1/e$ in the non-monotone matroid case and $1/2$ unconstrained. Our error is a deterministic multiplicative band on marginal gains, adversarial inside the band, so the answer is a finite ratio at every error level rather than either full recovery or vacuity, and the selection error we state it is a property of the finished run.

Learning-augmented algorithms and learn-then-optimize. This framework asks for algorithms that improve with better predictions but do not degrade much when the predictions are poor (Purohit et al., 2018), the first property being consistency in its learning-augmented sense; Lykouris & Vassilvitskii (2021) realize it for caching with a competitive ratio that decreases as the oracle’s error decreases and is capped at $O(\log k)$. Learning a submodular objective is itself hard, since PMAC-learning monotone submodular functions under arbitrary distributions forces approximation factor $\tilde{\Omega}(n^{1/3})$ (Balcan & Harvey, 2018), and against a background of negative results in which optimizing a learned surrogate lands arbitrarily far from the true optimum, Rosenfeld et al. (2018) characterize which surrogates can be optimized at all. Predicting other objects does help: a $1/2 - \epsilon$ expected ratio for dynamic monotone submodular maximization under predicted insertion and deletion times, with amortized update time degrading in the prediction error η_{AB} (Agarwal & Balkanski, 2024),¹ and noisy predictions of the optimal solution bypassing classical hardness for problems such as max-cut (Cohen-Addad et al., 2024). What is predicted there is the solution, the schedule, or a function learned once and then optimized; here the prediction is the objective, queried adaptively during the run, and the guarantee is stated in an error measure the run reports.

Optimization from samples and query models. When only sampled sets and values are available, bounded curvature still permits a positive result (Balkanski et al., 2016), while in general no constant-factor approximation follows from polynomially many samples even for coverage functions learnable to arbitrary accuracy (Balkanski et al., 2017). We keep adaptive query access and move the difficulty into the accuracy of the answers, so Theorem 12 concerns what a bounded number of small queries extracts from an inaccurate predictor, not what a fixed sample reveals.

¹Their η_{AB} counts elements whose actual insertion or deletion time is far from the predicted one; it is unrelated to the multiplicative marginal-gain error η of Definition 1 and is renamed here to keep the two apart.

4 THEORETICAL RESULTS

4.1 AN ERROR BOUND IS NECESSARY

Proposition 3 (No bound, no guarantee). *If no upper bound on η is assumed, then for every algorithm with arbitrary query access to $\tilde{f}$ and every $n \geq 2K$ there are pairs $(f, \tilde{f})$ on which the output T satisfies $f(T) \leq \frac{K}{n-K} f(O^*)$. Consequently no constant worst-case ratio is achievable without an error assumption.*

4.2 THE GUARANTEE, STATED FOR THE SELECTION ERROR

Proposition 4 (Guarantee of predictive greedy). *Let f be monotone submodular with $f(\emptyset) = 0$ and let $T = S^K$ be the output of a run of predictive greedy whose selection error is η^{sel} . Then, with $L_K(x) = 1 - (1 - \frac{1}{xK})^K$,*

$$f(T) \geq L_K(\eta^{\text{sel}}) f(O^*) \geq (1 - e^{-1/\eta^{\text{sel}}}) f(O^*).$$

The same bound therefore holds with η^{sel} replaced by η^{tr} or by η , and the three bounds are ordered $L_K(\eta^{\text{sel}}) \geq L_K(\eta^{\text{tr}}) \geq L_K(\eta)$.

Proposition 4 is restated in the prediction-error model of Section 2; it is essentially due to Goundan and Schulz (Goundan & Schulz, 2007, Theorem 1). A proof is included in Appendix B.2 for completeness.

Remark 5. η^{sel} is computable from the run when f becomes observable in hindsight, which makes Proposition 4 a certificate: the measured η^{sel} of a finished run certifies the printed bound for that run. The experiments of Section 5 report measured η^{sel} and the resulting certified bounds.

4.3 PER- K TIGHTNESS

Theorem 6 (Tightness for every K). *For every $K \geq 2$ and $\hat{a} > 1$ there is an instance $(f, \tilde{f})$ on $2K$ elements with $\eta^{\text{sel}} = \eta^{\text{tr}} = \hat{a}$ on the adversarial-tie run of predictive greedy, whose output value is exactly $L_K(\hat{a}) f(O^*)$. Hence Proposition 4 is an equality on this family for every K , not only in the limit.*

4.4 THE COHERENCE LEMMA

Lemma 7 (Coherence). *Let f be monotone, $S \subseteq N$, $e, e' \notin S$, and suppose $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$. Then*

$$(i) \ d_e(S \cup \{e'\}) \geq \frac{1}{\eta} d_{e'}(S \cup \{e'\}), \quad (ii) \ (1 - \frac{1}{\eta}) d_{e'}(S \cup \{e'\}) \geq d_{e'}(S) - d_e(S).$$

4.5 THE EXACT WORST CASE

For $K \geq 2$, $\eta \geq 1$, let $k_1 = (K-1)\eta + 1$, $q = (K-1)\eta/k_1$, and for $0 \leq j \leq K-1$

$$V_j(\eta) = 1 - q^j \left(1 - \frac{K-j}{K\eta}\right).$$

Theorem 8 (Exact worst-case ratio of predictive greedy). *For every $K \geq 2$ and $\eta \geq 1$, under adversarial tie breaking,*

$$\rho_K(\eta) = \min_{0 \leq j \leq K-1} V_j(\eta),$$

and the minimum is attained by V_j on the segment $\eta \in [K-j, K-j+1]$ (with $V_0 = 1/\eta$ on $[K, \infty)$), so the breakpoints are the integers $2, \dots, K$. In particular $\rho_K(\eta) = 1/\eta$ exactly when $\eta \geq K$, and for $K \in \{2, 3, 4\}$ the closed forms are $\rho_2 = \min\{\frac{1}{\eta}, \frac{3}{2(\eta+1)}\}$, $\rho_3 = \frac{16\eta+3}{3(2\eta+1)^2}$, $\frac{7}{3(2\eta+1)}$, $\frac{1}{\eta}$ on $[1, 2]$, $[2, 3]$, $[3, \infty)$, and $\rho_4 = \frac{135\eta^2+36\eta+4}{4(3\eta+1)^3}$, $\frac{21\eta+2}{2(3\eta+1)^2}$, $\frac{13}{4(3\eta+1)}$, $\frac{1}{\eta}$ on $[1, 2]$, $[2, 3]$, $[3, 4]$, $[4, \infty)$.

Remark 9. L_K and ρ_K differ by $O(1/K)$ for fixed η , and the previously used explicit family attains only $U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$, which is strictly above V_{K-1} for every $\eta > 1$. The predictor $\tilde{f}$ of the attaining instances is monotone but not submodular. If the surrogate is itself submodular,

the exact worst case strictly improves for $\eta < K - 1$: at $K = 3$, $\eta = 1.5$ the value rises from 9/16 to 19/33 and the family above becomes inadmissible, so U_K is no longer an upper bound in that model. A general characterization is left open.

4.6 THE $1/\eta$ CEILING

Theorem 10 (Ceiling at $1/\eta$). *For every deterministic algorithm with arbitrary query access to $\tilde{f}$, every $\eta_u, \eta_o \geq 1$, and every $n \geq 2K$, there is a pair $(f, \tilde{f})$ with error exactly (η_u, η_o) on which the output T satisfies $f(T) \leq f(O^*)/\eta$. For randomized algorithms the bound becomes $(1 - \frac{K}{n})\frac{1}{\eta} + \frac{K}{n}$ in expectation. Conversely, exhaustive search over all K -subsets of predicted values achieves $f(\hat{S}) \geq f(O^*)/\eta$ on every instance.*

4.7 ASYMPTOTICS

Corollary 11 (Limit in K). *For fixed $\eta \geq 1$, both $L_K(\eta)$ and $\rho_K(\eta)$ converge to $1 - e^{-1/\eta}$ as $K \rightarrow \infty$; for L_K the convergence is monotone in K .*

4.8 HARDNESS FOR BOUNDED-QUERY ALGORITHMS

The ceiling of Theorem 10 does not restrict the query budget. The final result shows that with polynomially many small queries no algorithm can beat the greedy guarantee by more than a vanishing margin. For $\theta \geq 1$ let $a_\theta = 1 - \frac{1}{\theta K}$, $1 + \delta(\theta) = \max\{a_\theta^\tau \frac{K}{K-\tau}, a_\theta^{1-\tau}\}^2$ and $\Phi(\theta) = \theta(1 + \delta(\theta))$.

Theorem 12 (Bounded-query hardness, deterministic). *Fix $c \geq 0$, put $\tau = c + 1$, and let $K \geq 2\tau$ and $\eta > 1$ satisfy $\Phi(1) \leq \eta$ (sufficient: $K \geq \tau(1 + \frac{2}{\ln \eta})$), so that $\hat{\eta} = \Phi^{-1}(\eta) \in [1, \eta]$ is well defined. Let $n \geq 4K^{c+2}$. For every deterministic algorithm making at most n^c queries to $\tilde{f}$, each on a set of size at most K , there is an instance $(f, \tilde{f})$ with monotone submodular f and single-element error exactly η on which the output T (of size at most K) satisfies*

$$f(T) \leq L_K(\hat{\eta}) f(O^*).$$

For randomized algorithms the same bound holds up to an additive $\varepsilon_n = \frac{K}{n} + \frac{K^{2c+4}}{(c+2)!n^2}$ in expectation. Moreover $K\delta(\eta) \rightarrow \max\{2\tau(1 - \frac{1}{\eta}), \frac{2(\tau-1)}{\eta}\}$ and $L_K(\hat{\eta}) \rightarrow 1 - e^{-1/\eta}$ as $K \rightarrow \infty$.

Remark 13. *Combined with Proposition 4, the bounded-query optimum at error level η is pinned to $1 - e^{-1/\eta}$ asymptotically in K : predictive greedy achieves $L_K(\eta) \rightarrow 1 - e^{-1/\eta}$ with K evaluations of $\tilde{f}$ per step, and no algorithm within the stated query budget exceeds $L_K(\hat{\eta}) \rightarrow 1 - e^{-1/\eta}$. At finite K a gap of order c/K between the two curves remains open.*

5 EXPERIMENTS

The experiments ask what Proposition 4 certifies on surrogates that were never built to satisfy Definition 1, and how far that certificate sits below what predictive greedy delivers. The certificate is read at the measured selection error η^{sel} of the finished run (Remark 5). Three surrogate families are ordered by their distance from the model of Section 2: surrogates *learned* from data, surrogates that evaluate the same objective on a *partially observed* input, and *heuristic* surrogates for a true objective that itself lies outside the model. The instances constructed for Theorems 8 and 6 are run through the same pipeline as a control.

Setup. In every family f is the quantity of interest, $\tilde{f}$ is what the algorithm may query, and nothing synthetic is inserted between them. (E1) *Feature selection*: $f(S)$ is the held-out accuracy of a decision tree trained on the features S , and $\tilde{f}(S)$ the mean 5-fold cross-validation accuracy of the same estimator on the training split alone, so the held-out split is structurally out of the predictor’s reach; 4 datasets (the largest with 25,375 rows), 30 splits each, $K = 1, \dots, 7$. (E2) *Influence maximization*: f is one-hop coverage in a network and $\tilde{f}$ the same coverage on an observed graph keeping each edge independently with probability $p \in \{0.3, 0.5, 0.8\}$ (Kempe et al., 2003); 4 networks (largest 50,515 nodes, 819,306 edges), 20 observed graphs per network and p , giving 240 trajectories of

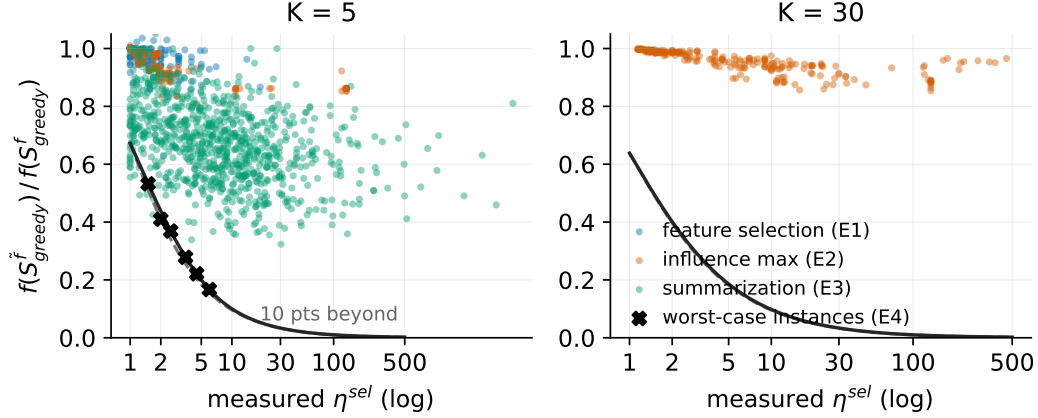


Figure 1: Measured ratio against measured selection error η^{sel} (log scale) at $K = 5$ and $K = 30$. Solid: the exact worst case ρ_K of Theorem 8; dashed: the bound $L_K(\eta^{\text{sel}})$ of Proposition 4. Colored points are runs of the three surrogate families, black crosses the constructed instances of Theorem 8 executed by the same pipeline, which were built at the budgets $K \in \{3, 5, 8\}$ and so appear in the left panel only. Real tasks sit far above both curves; the constructed instances sit on the solid one. Ratio denominators are greedy on the true objective, an upper-estimate proxy for the optimum.

length 30 (Rozemberczki et al., 2019; Leskovec et al., 2007; Yin et al., 2017). (E3) *Extractive summarization*: f is the ROUGE-1 F score against a reference summary and $\tilde{f}$ one of 3 standard submodular sentence-selection objectives that never see the reference (Lin & Bilmes, 2011); 299 articles in 3 categories, $K = 3, \dots, 7$. Ratios are $f(S^{\tilde{f}})/f(S^f)$, whose denominator is greedy on the true objective and hence an upper-estimate proxy for the optimum (Table 1, note (i)); on the one dataset small enough to enumerate the optimum, that proxy inflates a ratio by 1.8 %. Greedy uses cached lazy evaluation; accelerated variants carry guarantees of their own (Mirzasoleiman et al., 2015) but are not used, because measuring η^{sel} needs the exact per-step maximizer of the true gain. Appendix C gives the protocol, the baseline panel and the enumeration.

Learned surrogates. At $K = 7$, over 120 runs, the median ratio is 0.971 (interquartile range 0.943 to 0.998) at median selection error 2.0, so the median run certifies $L_7(\eta^{\text{sel}}) = 0.405$ against an exact worst case of $\rho_7 = 0.426$ at that same error; on the largest dataset alone the ratio is 0.999 at error 1.55, certifying 0.492. Which measure the guarantee is read at decides whether it says anything: the trajectory error of the same runs has median at least 19.4 (the reported figure discards candidate pairs whose gains fall below one quantization unit of the accuracy, so it is a lower estimate; Appendix C), and at that value Proposition 4 certifies only 0.050.

Partially observed surrogates. Here f and $\tilde{f}$ are one function of two nested edge sets, so the predictor errs only by missing coverage. At $K = 30$ the median ratio is 0.963 (interquartile range 0.936 to 0.989) at median selection error 4.3, certifying $L_{30}(\eta^{\text{sel}}) = 0.207$ against an exact worst case of 0.211. Only 0.0 % of steps have a nonpositive chosen true gain, and no predicted and true gain can disagree in sign, both by construction rather than by luck (Table 1, note (iii)).

Observation probability against error. This family lets the amount of error be dialled. Lowering p from 0.8 to 0.3 raises the median η^{sel} monotonically on all 4 networks, by factors from 3.2 to 88; the realized ratio degrades far more slowly over the same range, also monotonically on all 4 networks, from no worse than 0.988 down to between 0.882 and 0.963. The certified bound moves fastest of the three: $L_{30}(\eta^{\text{sel}})$ falls from the range 0.361 to 0.573 down to 0.008 to 0.230, by up to a factor 64 on one network. Missing observations therefore cost much less realized value than they cost certificate.

Heuristic surrogates outside the model. At $K = 5$, over 879 runs, the median ratio falls to 0.670 (interquartile range 0.576 to 0.757) at median selection error 7.2, where $L_5(\eta^{\text{sel}}) = 0.132$ and the

Table 1: Predictive greedy across three surrogate families. Medians over all runs at the stated K (IQR in brackets). L_K is the certified lower bound $L_K(\eta^{\text{sel}})$ of Proposition 4 evaluated at the median measured η^{sel} ; $d_t \leq 0$ % is the share of trajectory steps whose chosen true gain is non-positive.

Task (K)	Surrogate $\tilde{f}$	ratio	IQR	η^{sel}	L_K	sign-viol. %	$d_t \leq 0$ %
Feature sel. (7)	5-fold CV acc.	0.971	[0.943, 0.998]	2.0	0.405	22.7	21.4
Influence max. (30)	observed graph	0.963	[0.936, 0.989]	4.3	0.207	–	0.0 ^a
Summarization (5)	cover./div./FL	0.670	[0.576, 0.757]	7.2	0.132	10.0	20.0
Worst-case V_j (5)	constructed	0.278 ^a	exact	3.5	ρ_K	0	0.0

representative point $j = 2$, $\eta = 3.5$; every constructed instance realizes its theoretical value to 10^{-10} (results/E4_worst-case.csv).

Note (i): the ratio’s denominator is greedy-on- $\tilde{f}$, an upper-estimate proxy for OPT, so every ratio in this table is an upper estimate of $f(S^{\tilde{f}})/f(\text{OPT})$.

Note (ii): η^{sel} is defined on the steps with positive true gain, so the certified bound in the L_K column is a statement about those steps; the last column reports the share of steps with $d_t \leq 0$ (column `frac_steps_nonpos` of the row CSVs), which are outside that scope.

Note (iii): for influence maximization f and $\tilde{f}$ are coverage functions on nested edge sets, so opposite-sign $(d, \tilde{d})$ pairs cannot occur; the entry is “–” rather than a measured zero.

exact worst case is 0.140; the best single surrogate, the coverage objective, reaches 0.712. Across the three families the median selection error grows and the median ratio falls together, which is the spectrum Table 1 summarizes.

An out-of-model finding. The true objective of the third family does not satisfy the hypotheses of Proposition 4, and we measure by how much rather than leaving it implicit. On 70,560 triples $(A, A \cup \{b\}, e)$ drawn from the articles with $|A| \leq 3$, the ROUGE-1 F score violates submodularity on 2.14 % and monotonicity on 7.12 % of them, the largest submodularity violation being 0.020 in units of the score, while the 3 surrogates record 0 violations of either property on the same triples; a further 20.0 % of trajectory steps have a nonpositive chosen true gain and fall outside the scope of η^{sel} (Table 1, note (ii)). For this family the L_K column is therefore a reference value computed by the same formula and not a certificate, and recovering one would need a guarantee stated for approximately submodular objectives (Section 3).

The worst case, executed by the same code. 19 constructed instances were run through the pipeline that produced every other point: 16 instances of the V_j family of Theorem 8, one per segment at $K \in \{3, 5, 8\}$, and 3 of the per- K family of Theorem 6. Realized ratios match the theoretical values to 1.1×10^{-16} , and on the V_j instances the measured selection error matches the design error to 5.3×10^{-15} . The worst case is thus not an asymptotic figure of speech: it occurs point by point in the same implementation of greedy that produced the scatter, so the vertical distance in Figure 1 between the crosses and the colored points at a common η^{sel} is a distance between two behaviours of one algorithm, not between two measurement paths. Three auxiliary figures, the distribution of η^{sel} by budget, the observation probability sweep, and the true against predicted gains with an error band drawn in, are deferred to Appendix C.

6 CONCLUSION

AI USE STATEMENT

In this work, we used generative AI tools as research assistants: to write and run the LP, symbolic, and exhaustive verification scripts that back the status annotations of the theorems, to implement and execute the experiment pipelines, to draft theorem statements and proof text, and to organize data and references. We have reviewed all AI-assisted work: the authors re-ran the verification scripts and checked the proofs line by line before submission, and statements whose proofs have not been so checked are explicitly marked in the source with their verification status. No claim in this paper

rests solely on an unverified AI-generated argument. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

All numerical claims are backed by one-command scripts in the anonymous repository accompanying this submission (link withheld in this draft; an anonymized link is to be added at submission time): the factor-revealing and reduced LPs, the symbolic verifications (sympy, exact arithmetic), the worst-case instance realizations, and the three experiment pipelines with fixed seed lists. The appendix states, for each theorem, which parts are verified by which script. Data handling is documented in the repository inventory: the airline table and the BBC corpus ship with the repository; the reference summaries missing for two BBC categories are backfilled from a public CSV of the same dataset, with a token-level cross-validation of the backfill on the category present locally; the social graphs are public SNAP datasets used as replacements for graphs no longer available, as documented in the inventory. Experiment tables state that ratio denominators use the true-value greedy as an OPT proxy, and every reported lower bound states the error measure it is evaluated at.

REFERENCES

- Arpit Agarwal and Eric Balkanski. Learning-augmented dynamic submodular maximization. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/19cdab1dee61d55158cf106244ceab08-Abstract-Conference.html.
- Maria-Florina Balcan and Nicholas J. A. Harvey. Submodular functions: learnability, structure, and optimization. *SIAM Journal on Computing*, 47(3):703–754, 2018. doi: 10.1137/120888909.
- Eric Balkanski, Aviad Rubinstein, and Yaron Singer. The power of optimization from samples. In *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pp. 4017–4025, 2016. URL <https://proceedings.neurips.cc/paper/2016/hash/c8758b517083196f05ac29810b924aca-Abstract.html>.
- Eric Balkanski, Aviad Rubinstein, and Yaron Singer. The limitations of optimization from samples. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2017)*, pp. 1016–1027, 2017. doi: 10.1145/3055399.3055406.
- Kshipra Bhawalkar, Yang Cai, Zhe Feng, Christopher Liaw, and Tao Lin. A unified approach to submodular maximization under noise. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, 2025. URL http://papers.nips.cc/paper_files/paper/2025/hash/f3064f7a0ca2328ecb41a3aef6177d68-Abstract-Conference.html.
- Vincent Cohen-Addad, Tommaso d’Orsi, Anupam Gupta, Euiwoong Lee, and Debmalya Panigrahi. Learning-augmented approximation algorithms for maximum cut and related problems. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL http://papers.nips.cc/paper_files/paper/2024/hash/2db08b94565c0d582cc53de6cee5fd47-Abstract-Conference.html.
- Abhimanyu Das and David Kempe. Submodular meets spectral: greedy algorithms for subset selection, sparse approximation and dictionary selection. In *Proceedings of the 28th International Conference on Machine Learning (ICML 2011)*, pp. 1057–1064, 2011. URL https://icml.cc/2011/papers/542_icmlpaper.pdf.
- Ethan R. Elenberg, Rajiv Khanna, Alexandros G. Dimakis, and Sahand Negahban. Restricted strong convexity implies weak submodularity. *The Annals of Statistics*, 46(6B):3539–3568, 2018. doi: 10.1214/17-AOS1679.
- Uriel Feige. A threshold of $\ln n$ for approximating set cover. *Journal of the ACM*, 45(4):634–652, 1998. doi: 10.1145/285055.285059.
- Pranava R. Goundan and Andreas S. Schulz. Revisiting the greedy approach to submodular set function maximization. Working paper, Massachusetts Institute of Technology, 2007. URL <https://optimization-online.org/2007/08/1740/>. Optimization Online preprint 1740.

- Avinatan Hassidim and Yaron Singer. Submodular optimization under noise. In Satyen Kale and Ohad Shamir (eds.), *Proceedings of the 2017 Conference on Learning Theory (COLT)*, volume 65 of *Proceedings of Machine Learning Research*, pp. 1069–1122. PMLR, 2017. URL <https://proceedings.mlr.press/v65/hassidim17a.html>.
- Thibaut Horel and Yaron Singer. Maximization of approximately submodular functions. In *Advances in Neural Information Processing Systems 29 (NIPS 2016)*, pp. 3045–3053, 2016. URL <https://proceedings.neurips.cc/paper/2016/hash/81c8727c62e800be708dbf37c4695dff-Abstract.html>.
- David Kempe, Jon Kleinberg, and Éva Tardos. Maximizing the spread of influence through a social network. In *Proceedings of the 9th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2003)*, pp. 137–146, 2003. doi: 10.1145/956750.956769.
- Jure Leskovec, Jon Kleinberg, and Christos Faloutsos. Graph evolution: densification and shrinking diameters. *ACM Transactions on Knowledge Discovery from Data*, 1(1):2, 2007. doi: 10.1145/1217299.1217301.
- Hui Lin and Jeff Bilmes. A class of submodular functions for document summarization. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pp. 510–520, Portland, Oregon, USA, 2011. Association for Computational Linguistics. URL <https://aclanthology.org/P11-1052/>.
- Thodoris Lykouris and Sergei Vassilvitskii. Competitive caching with machine learned advice. *Journal of the ACM*, 68(4):24:1–24:25, 2021. doi: 10.1145/3447579.
- Baharan Mirzasoleiman, Ashwinkumar Badanidiyuru, Amin Karbasi, Jan Vondrák, and Andreas Krause. Lazier than lazy greedy. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence (AAAI 2015)*, pp. 1812–1818, 2015. doi: 10.1609/aaai.v29i1.9486.
- George L. Nemhauser and Laurence A. Wolsey. Best algorithms for approximating the maximum of a submodular set function. *Mathematics of Operations Research*, 3(3):177–188, 1978. doi: 10.1287/moor.3.3.177.
- George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978. doi: 10.1007/BF01588971.
- Manish Purohit, Zoya Svitkina, and Ravi Kumar. Improving online algorithms via ML predictions. In *Advances in Neural Information Processing Systems 31 (NeurIPS 2018)*, pp. 9684–9693, 2018. URL <https://proceedings.neurips.cc/paper/2018/hash/73a427badebe0e32caa2e1fc7530b7f3-Abstract.html>.
- Nir Rosenfeld, Eric Balkanski, Amir Globerson, and Yaron Singer. Learning to optimize combinatorial functions. In Jennifer Dy and Andreas Krause (eds.), *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, volume 80 of *Proceedings of Machine Learning Research*, pp. 4374–4383. PMLR, 2018. URL <https://proceedings.mlr.press/v80/rosenfeld18a.html>.
- Benedek Rozemberczki, Ryan Davies, Rik Sarkar, and Charles Sutton. GEMSEC: graph embedding with self clustering. In *Proceedings of the 2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2019)*, pp. 65–72, 2019. doi: 10.1145/3341161.3342890.
- Hao Yin, Austin R. Benson, Jure Leskovec, and David F. Gleich. Local higher-order graph clustering. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2017)*, pp. 555–564, 2017. doi: 10.1145/3097983.3098069.

Table 2: Notation used throughout the paper.

N, n	ground set and its size
K	cardinality budget
f	monotone submodular objective, $f(\emptyset) = 0$; not queryable
$\tilde{f}$	predictor of f ; the only queryable object
$d_e(S), \tilde{d}_e(S)$	true and predicted marginal gain of e at S
η_u, η_o	under- and over-prediction factors ($d/\eta_u \leq \tilde{d} \leq \eta_o d$ on single-element gains)
$\eta = \eta_u \eta_o$	scalar (global) prediction error
η^{sel}	selection error of a run (Definition 2); main text uses this
η^{tr}	trajectory error (error bound restricted to the run’s states); one remark only
O^*, F^{OPT}	an optimal K -set and its value
$\alpha \in (0, 1]$	approximation ratio convention $F^{\text{ALG}} \geq \alpha F^{\text{OPT}}$
$\rho_K(\eta)$	exact worst-case ratio of predictive greedy, adversarial ties
$L_K(x) = 1 - (1 - \frac{1}{xK})^K$	guarantee curve (Proposition 4)
$k_1 = (K-1)\eta + 1, q = (K-1)\eta/k_1$	segment parameters of the exact worst case
$V_j(\eta) = 1 - q^j(1 - \frac{K-j}{K\eta})$	segment values; $\rho_K = \min_j V_j$ (Theorem 8)
$U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$	value of the per- K tightness family at global error η
$a_\theta, \delta(\theta), \Phi, \hat{\eta}$	hardness-family parameters (Section 4.8)
$\tau = c + 1, Q = n^c$	balancedness margin and query budget of the hardness theorem
ratio (experiments)	$f(S_{\text{greedy}}^{\tilde{f}})/f(S_{\text{greedy}}^f)$; the denominator is an OPT proxy

A NOTATION

B PROOFS

Three conventions are in force throughout this appendix. First, f is monotone submodular with $f(\emptyset) = 0$ and $\tilde{f}(\emptyset) = 0$, and submodularity is used in its diminishing-returns form: $d_e(T) \leq d_e(S)$ for $S \subseteq T$ and $e \notin T$. Second, an optimal set is written O^* , $|O^*| \leq K$, and instances are normalized so that $f(O^*) = 1$ whenever a ratio is computed. Third, the tie-breaking rule of a greedy run is part of the instance: “adversarial ties” means that when several elements maximize the predicted gain, the one named by the construction is taken.

Two of the constructions below are counting functions, so the following observation is recorded once and used twice.

Lemma 14 (counting functions). *Let $N = N_1 \cup \dots \cup N_r$ be a partition and let $h(S) = H(|S \cap N_1|, \dots, |S \cap N_r|)$ for a function H on the box $\prod_i \{0, 1, \dots, |N_i|\}$. Write $\Delta_i H(v) = H(v + u_i) - H(v)$ for the forward difference in coordinate i (u_i the i th unit vector), defined when $v_i < |N_i|$. If $\Delta_i H \geq 0$ for every i and every v , and $\Delta_\ell \Delta_i H \leq 0$ for every pair (i, ℓ) , including $i = \ell$, and every v at which both differences are defined, then h is monotone and submodular.*

Argument. Let $v(S)$ denote the count vector of S . For $e \in N_i$ with $e \notin S$ one has $d_e(S) = \Delta_i H(v(S))$, which is nonnegative by hypothesis, so h is monotone. For $S \subseteq T$ and $e \in N_i \setminus T$, the vectors satisfy $v(S) \leq v(T)$ coordinatewise and $v(T)_i < |N_i|$. Walk from $v(S)$ to $v(T)$ by single unit steps, raising coordinate i last: at each step the quantity $\Delta_i H$ changes by one second difference $\Delta_\ell \Delta_i H \leq 0$, and every intermediate vector has i th coordinate at most $v(T)_i < |N_i|$, so $\Delta_i H$ is defined all along. Hence $d_e(T) = \Delta_i H(v(T)) \leq \Delta_i H(v(S)) = d_e(S)$, which is the diminishing-returns form of submodularity.

B.1 NECESSITY OF AN ERROR BOUND (PROPOSITION 3)

Fix $K \geq 1$ and $n \geq 2K$, and let $\mathcal{A}$ be an algorithm with arbitrary query access to $\tilde{f}$. Put

$$\gamma = \frac{K^2}{n(n-K)} \in (0, 1], \quad \tilde{f}(S) = |S|,$$

where $\gamma \leq 1$ because $n \geq 2K$ gives $n(n-K) \geq 2K \cdot K > K^2$. For a K -set $O \subseteq N$ define

$$f_O(S) = |S \cap O| + \gamma |S \setminus O|.$$

The pair is admissible and has finite error. f_O is modular with coefficients in $\{1, \gamma\} \subseteq (0, 1]$, so it is monotone and submodular with $f_O(\emptyset) = 0$, and $\tilde{f}(\emptyset) = 0$. Every single-element gain is $\tilde{d}_e(S) = 1$ and $d_e(S) \in \{1, \gamma\}$, so $\eta_u = \max_{S,e} d/\tilde{d} = 1$ and $\eta_o = \max_{S,e} \tilde{d}/d = 1/\gamma$: the error is exactly $\eta = 1/\gamma = n(n-K)/K^2$, which is finite for every n and grows without bound as $n \rightarrow \infty$. Also $\max_{|S| \leq K} f_O(S) = f_O(O) = K$, because $\gamma \leq 1$ makes the K largest modular coefficients exactly those of O .

Deterministic algorithms. $\tilde{f}$ does not depend on O , so the whole query transcript of $\mathcal{A}$, and therefore its output T with $|T| \leq K$, are the same for every O . Since $n \geq 2K$ and $|T| \leq K$, a K -set $O \subseteq N \setminus T$ exists. For that O , $T \cap O = \emptyset$ gives

$$\frac{f_O(T)}{f_O(O^*)} = \frac{\gamma|T|}{K} \leq \gamma = \frac{K^2}{n(n-K)} \leq \frac{K}{n-K}.$$

Randomized algorithms. Let O be a uniformly random K -subset, independent of the random string of $\mathcal{A}$. The transcript still does not depend on O , so the output T is independent of O , and the hypergeometric mean gives $\mathbb{E}[|T \cap O|] = \mathbb{E}|T| \cdot K/n \leq K^2/n$. Bounding $f_O(T) \leq |T \cap O| + \gamma K$ and dividing by $f_O(O^*) = K$,

$$\mathbb{E}\left[\frac{f_O(T)}{f_O(O^*)}\right] \leq \frac{K}{n} + \gamma = \frac{K}{n} + \frac{K^2}{n(n-K)} = \frac{K(n-K) + K^2}{n(n-K)} = \frac{K}{n-K},$$

and $\min_O \leq \mathbb{E}_O$ produces a single O attaining the bound in expectation over the random string. Letting $n \rightarrow \infty$ with K fixed drives the right side to 0, so no constant worst-case ratio survives when η is unconstrained.

No script checks the computations of this proof.

B.2 THE GUARANTEE (PROPOSITION 4)

This proof is included for completeness; the statement is essentially Theorem 1 of Goundan & Schulz (2007), restated in the prediction-error model of Section 2.

Let $S^0 = \emptyset, S^1, \dots, S^K$ be the states of a run of predictive greedy with picks $e_0, \dots, e_{K-1}$ and selection error η^{sel} , and put $r_t = f(O^*) - f(S^t)$.

Step 1: one greedy step covers a $1/(\eta^{\text{sel}} K)$ share of the residual. Let $t < K$ be a step with $d_{e_t}(S^t) > 0$. Enumerate $O^* \setminus S^t = \{o_1, \dots, o_m\}$, $m \leq |O^*| \leq K$, and telescope:

$$f(O^* \cup S^t) - f(S^t) = \sum_{i=1}^m \left[f(S^t \cup \{o_1, \dots, o_i\}) - f(S^t \cup \{o_1, \dots, o_{i-1}\}) \right] \leq \sum_{i=1}^m d_{o_i}(S^t),$$

one application of submodularity per term. Monotonicity gives $f(O^* \cup S^t) \geq f(O^*)$, so $r_t \leq \sum_{i=1}^m d_{o_i}(S^t) \leq m \max_{e \notin S^t} d_e(S^t) \leq K \max_{e \notin S^t} d_e(S^t)$. This is the only place where the count $m \leq K$ enters. Definition 2 applied at this step reads $\max_{e \notin S^t} d_e(S^t) \leq \eta^{\text{sel}} d_{e_t}(S^t)$, hence

$$d_{e_t}(S^t) \geq \frac{1}{\eta^{\text{sel}} K} r_t, \quad \text{and so} \quad r_{t+1} = r_t - d_{e_t}(S^t) \leq \left(1 - \frac{1}{\eta^{\text{sel}} K}\right) r_t.$$

Step 2: why only the steps with positive chosen gain matter. Monotonicity of f makes $d_{e_t}(S^t) \geq 0$ always, so the excluded steps are exactly those with $d_{e_t}(S^t) = 0$. Suppose t is the first such step. By Definition 1, $\tilde{d}_{e_t}(S^t) \leq \eta_o d_{e_t}(S^t) = 0$, and e_t maximizes the predicted gain, so $\tilde{d}_e(S^t) = 0$ for every $e \notin S^t$; the lower band $d_e(S^t) \leq \eta_u \tilde{d}_e(S^t)$ then forces $d_e(S^t) = 0$ for every $e \notin S^t$. The covering inequality $r_t \leq K \max_{e \notin S^t} d_e(S^t)$ of Step 1, whose derivation does not use positivity of the chosen gain, then reads $r_t \leq K \cdot 0 = 0$, that is, $f(S^t) \geq f(O^*)$: the run has already reached the optimal value and $r_s = 0$ for every $s \geq t$. Therefore the contraction of Step 1 may be applied at every step, treating the residual as 0 from the first nonpositive step onwards.

Step 3: unrolling. With $r_0 = f(O^*)$, iterating K times gives $r_K \leq (1 - \frac{1}{\eta^{\text{sel}} K})^K f(O^*)$ and hence

$$f(S^K) = f(O^*) - r_K \geq \left(1 - \left(1 - \frac{1}{\eta^{\text{sel}} K}\right)^K\right) f(O^*) = L_K(\eta^{\text{sel}}) f(O^*).$$

Since $\eta^{\text{sel}} \geq 1$ and $K \geq 1$, the quantity $u = 1/(\eta^{\text{sel}} K)$ lies in $(0, 1]$, and $1 - u \leq e^{-u}$ gives $L_K(\eta^{\text{sel}}) \geq 1 - e^{-1/\eta^{\text{sel}}}$.

Step 4: the three error measures. At every state S^t of the run,

$$\max_{e \notin S^t} d_e(S^t) \leq \eta_u \max_{e \notin S^t} \tilde{d}_e(S^t) = \eta_u \tilde{d}_{e_t}(S^t) \leq \eta_u \eta_o d_{e_t}(S^t),$$

the first step by the lower band applied to the maximizing element, the middle equality by the greedy choice, the last by the upper band at (S^t, e_t) . Reading the two ends as a step ratio of Definition 2 gives $\eta^{\text{sel}} \leq \eta_u \eta_o$ when the band is the global one, and $\eta^{\text{sel}} \leq \eta^{\text{tr}}$ when it is restricted to the states of the run; $\eta^{\text{tr}} \leq \eta$ holds because the states of the run are a subset of all sets. The map $x \mapsto 1 - \frac{1}{xK}$ is increasing on $x \geq 1$ with values in $[0, 1]$, so $x \mapsto (1 - \frac{1}{xK})^K$ is increasing and L_K is decreasing. Hence $L_K(\eta^{\text{sel}}) \geq L_K(\eta^{\text{tr}}) \geq L_K(\eta)$, and the bound of Step 3 holds a fortiori with η^{tr} or η in place of η^{sel} .

No script certifies the steps of this argument. That the bound is met with equality, and that the three rulers read $\eta^{\text{sel}} = \eta^{\text{tr}} = \hat{a} < \eta$ on the family of Appendix B.3, is machine-checked by `results/F2_etaset_tight.py`.

B.3 PER- K TIGHTNESS (THEOREM 6)

Fix $K \geq 2$ and $\hat{a} > 1$, and put

$$a = 1 - \frac{1}{\hat{a}K} \in (0, 1), \quad N = B \cup O, \quad |B| = |O| = K, \quad n = 2K.$$

For $S \subseteq N$ write $x = |S \cap B|$ and $y = |S \cap O|$, and define the pair $(f, \tilde{f})$ by

$$\begin{aligned} f(S) &= F(x, y) = 1 - a^x \left(1 - \frac{y}{K}\right), \\ \tilde{f}(S) &= G(x, y) = \begin{cases} 1 - a^x, & y = 0, \\ 1 - a^x + a^x \left[(1 - a) + \frac{(y - 1)a}{K - 1}\right], & 1 \leq y \leq K. \end{cases} \end{aligned} \quad (1)$$

Both are counting functions of (x, y) , so Lemma 14 applies with $r = 2$, $N_1 = B$, $N_2 = O$. Note $F(0, 0) = 0$ and $G(0, 0) = 0$.

Step 1: a closed form for the second branch of G . For $1 \leq y \leq K$,

$$\begin{aligned} 1 - a^x + a^x(1 - a) + \frac{(y - 1)a^{x+1}}{K - 1} &= 1 - a^{x+1} + \frac{(y - 1)a^{x+1}}{K - 1} \\ &= 1 - a^{x+1} \frac{(K - 1) - (y - 1)}{K - 1} = 1 - a^{x+1} \frac{K - y}{K - 1}. \end{aligned}$$

In particular $G(x, K) = 1$ for every x , and the two branches agree at $y = 1$: the second branch gives $1 - a^{x+1}$, which is $G(x, 0) + a^x(1 - a)$.

Step 2: the four gain identities. Writing Δ_x and Δ_y for the forward differences in x and y , the four cases are computed directly from equation 1 and Step 1:

$$\Delta_x F(x, y) = a^x(1 - a) \frac{K - y}{K}, \quad \Delta_y F(x, y) = \frac{a^x}{K}, \quad (2)$$

$$\Delta_x G(x, 0) = a^x(1 - a), \quad \Delta_y G(x, 0) = a^x(1 - a), \quad (3)$$

$$\Delta_x G(x, y) = a^{x+1}(1 - a) \frac{K - y}{K - 1}, \quad \Delta_y G(x, y) = \frac{a^{x+1}}{K - 1}, \quad (4)$$

the last row for $1 \leq y \leq K$ in the x direction and for $1 \leq y \leq K - 1$ in the y direction. The two entries of equation 3 are the same number because both read $(1 - a^{x+1}) - (1 - a^x)$: the first crosses no branch, and the second crosses from $G(x, 0)$ to $G(x, 1) = 1 - a^{x+1}$ by Step 1. Dividing equation 3 and equation 4 by the corresponding entries of equation 2 gives the four ratios, with $R := aK/(K - 1)$:

$$\frac{\Delta_x G(x, 0)}{\Delta_x F(x, 0)} = 1, \quad \frac{\Delta_y G(x, 0)}{\Delta_y F(x, 0)} = \frac{a^x(1 - a)}{a^x/K} = K(1 - a) = \frac{1}{\hat{a}},$$

and, in the range $1 \leq y \leq K - 1$ where both are defined,

$$\frac{\Delta_x G(x, y)}{\Delta_x F(x, y)} = \frac{a^{x+1}(1 - a)(K - y)/(K - 1)}{a^x(1 - a)(K - y)/K} = R, \quad \frac{\Delta_y G(x, y)}{\Delta_y F(x, y)} = \frac{a^{x+1}/(K - 1)}{a^x/K} = R,$$

where $K(1 - a) = 1/\hat{a}$ is the definition of $\hat{a}$. At $y = K$ the only remaining direction is x , and both gains vanish: $\Delta_x F(x, K) = 0$ by equation 2 and $\Delta_x G(x, K) = 0$ by Step 1. So a zero true gain always comes with a zero predicted gain, as Definition 1 demands.

Step 3: monotonicity and submodularity of f , case by case. The two first differences in equation 2 are nonnegative for $0 \leq y \leq K$ and $x \geq 0$, since $a \in (0, 1)$. The four second differences are

$$\begin{aligned} \Delta_x \Delta_x F &= -a^x(1 - a)^2 \frac{K - y}{K} \leq 0, & \Delta_y \Delta_x F &= -\frac{a^x(1 - a)}{K} \leq 0, \\ \Delta_x \Delta_y F &= \frac{a^{x+1} - a^x}{K} = -\frac{a^x(1 - a)}{K} \leq 0, & \Delta_y \Delta_y F &= \frac{a^x}{K} - \frac{a^x}{K} = 0 \leq 0. \end{aligned}$$

The two mixed differences coincide, as they must. Lemma 14 now gives that f is monotone and submodular on the whole of 2^N . (No claim is made about $\tilde{f}$; it is monotone but not submodular, which the model of Section 2 permits.)

Step 4: the global error of the pair. By Step 2 the set of single-element ratios $\tilde{d}_e(S)/d_e(S)$, over all S and all $e \notin S$ with $d_e(S) > 0$, is exactly $\{1, R, 1/\hat{a}\}$. Two sign computations order it:

$$R - 1 = \frac{aK - (K - 1)}{K - 1} = \frac{K - \frac{1}{\hat{a}} - K + 1}{K - 1} = \frac{1 - 1/\hat{a}}{K - 1} > 0, \quad 1 - \frac{1}{\hat{a}} = \frac{\hat{a} - 1}{\hat{a}} > 0,$$

both strict because $\hat{a} > 1$. Hence $\eta_o = \max\{1, R, 1/\hat{a}\} = R$ and $\eta_u = \max\{1, 1/R, \hat{a}\} = \hat{a}$, both attained, and

$$\eta = \eta_u \eta_o = \hat{a} \frac{aK}{K - 1} = \frac{K(\hat{a} - \frac{1}{K})}{K - 1} = \frac{K\hat{a} - 1}{K - 1}, \quad (5)$$

using $\hat{a}a = \hat{a} - 1/K$.

Step 5: the optimum. $1 - F(x, y) = a^x(K - y)/K \geq 0$, so $F \leq 1$ everywhere, and $F(0, K) = 1$. Since $|O| = K$, the value $f(O^*) = 1$ is attained at O , and $O^* = O$ may be taken.

Step 6: the greedy induction under adversarial ties. Claim: for $0 \leq t \leq K$ the state after t steps satisfies $S^t \subseteq B$ and $|S^t| = t$, that is $(x, y) = (t, 0)$. For $t = 0$ this is $S^0 = \emptyset$. Assume it at $t < K$. At the state $(t, 0)$ every candidate is either a remaining element of B , whose predicted gain is $\Delta_x G(t, 0) = a^t(1 - a)$, or an element of O , whose predicted gain is $\Delta_y G(t, 0) = a^t(1 - a)$; these are equal by equation 3, so every candidate is a maximizer and the step is a tie among all $2K - t$ of them. Adversarial tie breaking selects an element of B (which is possible because $t < K = |B|$), so S^{t+1} has $(x, y) = (t + 1, 0)$. After K steps $S^K = B$ and

$$f(S^K) = F(K, 0) = 1 - a^K = 1 - \left(1 - \frac{1}{\hat{a}K}\right)^K = L_K(\hat{a}),$$

which is $L_K(\hat{a})f(O^*)$ by Step 5.

Step 7: the two run-dependent errors equal $\hat{a}$. At the state $S^t = (t, 0)$ the true gains are $a^t(1 - a) = a^t/(\hat{a}K)$ for an element of B and a^t/K for an element of O , by equation 2. Since $\hat{a} > 1$, the largest true gain at that state is a^t/K while the chosen element has gain $a^t/(\hat{a}K) > 0$, so the step ratio of Definition 2 is

$$\frac{\max_{e \notin S^t} d_e(S^t)}{d_{e_t}(S^t)} = \frac{a^t/K}{a^t/(\hat{a}K)} = \hat{a}$$

at every one of the K steps, so $\eta^{\text{sel}} = \hat{a}$. For η^{tr} , only the states of the run matter, and at those states $y = 0$, so by Step 2 the only two ratios that occur are 1 (direction x) and $1/\hat{a}$ (direction y); therefore the trajectory-restricted factors are $\eta_o^{\text{tr}} = \max\{1, 1/\hat{a}\} = 1$ and $\eta_u^{\text{tr}} = \max\{1, \hat{a}\} = \hat{a}$, so $\eta^{\text{tr}} = \hat{a}$ as well. Comparing with equation 5, $\eta - \hat{a} = \frac{K\hat{a} - 1 - (K-1)\hat{a}}{K-1} = \frac{\hat{a}-1}{K-1} > 0$, so on this family the selection and trajectory rulers agree and are strictly smaller than the global one.

Step 8: the same family read at fixed global error. Solving equation 5 for $\hat{a}$ gives $\hat{a} = (\eta(K - 1) + 1)/K$, hence $a = 1 - \frac{1}{\hat{a}K} = 1 - \frac{1}{\eta(K-1)+1}$ and

$$f(S^K) = 1 - a^K = 1 - \left(1 - \frac{1}{\eta(K-1)+1}\right)^K = U_K(\eta).$$

So the same instances that make Proposition 4 an equality at $\eta^{\text{sel}} = \hat{a}$ realize only U_K when charged the global error η , which is the comparison recorded in Remark 9.

Steps 1 to 8 give Theorem 6: for every $K \geq 2$ and every $\hat{a} > 1$ the pair equation 1 on $2K$ elements has monotone submodular f , finite error, an adversarial-tie greedy run with $\eta^{\text{sel}} = \eta^{\text{tr}} = \hat{a}$, and output value exactly $L_K(\hat{a})f(O^*)$.

The identities of this proof are machine-checked by `results/T5_symbolic.py` and `results/F2_etaset_tight.py`.

B.4 COHERENCE (LEMMA 7)

Let $S \subseteq N$ and $e, e' \notin S$ with $e \neq e'$, and assume $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$.

The exchange identity. Expanding the same quantity $\tilde{f}(S \cup \{e, e'\}) - \tilde{f}(S)$ in the two possible orders, first e then e' , and first e' then e :

$$\tilde{d}_e(S) + \tilde{d}_{e'}(S \cup \{e\}) = \tilde{f}(S \cup \{e, e'\}) - \tilde{f}(S) = \tilde{d}_{e'}(S) + \tilde{d}_e(S \cup \{e'\}).$$

This uses nothing but the definition of a set function. Subtracting and applying the hypothesis $\tilde{d}_e(S) - \tilde{d}_{e'}(S) \geq 0$,

$$\tilde{d}_{e'}(S \cup \{e\}) - \tilde{d}_e(S \cup \{e'\}) = \tilde{d}_{e'}(S) - \tilde{d}_e(S) \leq 0. \quad (6)$$

Part (i). Chain the two bands of Definition 1 around equation 6:

$d_{e'}(S \cup \{e\}) \leq \eta_u \tilde{d}_{e'}(S \cup \{e\}) \leq \eta_u \tilde{d}_e(S \cup \{e'\}) \leq \eta_u \eta_o d_e(S \cup \{e'\}) = \eta d_e(S \cup \{e'\})$, the first inequality by the lower band at $(S \cup \{e\}, e')$, the second by equation 6, the third by the upper band at $(S \cup \{e'\}, e)$. Dividing by η gives (i).

Part (ii). The same two-order expansion applied to f reads $d_e(S) + d_{e'}(S \cup \{e\}) = d_{e'}(S) + d_e(S \cup \{e'\})$, that is

$$d_{e'}(S) - d_e(S) = d_{e'}(S \cup \{e\}) - d_e(S \cup \{e'\}).$$

Substituting part (i) in the form $-d_e(S \cup \{e'\}) \leq -\frac{1}{\eta} d_{e'}(S \cup \{e\})$,

$$d_{e'}(S) - d_e(S) \leq d_{e'}(S \cup \{e\}) - \frac{1}{\eta} d_{e'}(S \cup \{e\}) = \left(1 - \frac{1}{\eta}\right) d_{e'}(S \cup \{e\}),$$

which is (ii). Only $\eta \geq 1$ is used beyond the bands, and only to keep the factor $1 - \frac{1}{\eta}$ nonnegative when the inequality is applied later.

No script checks the computations of this proof. The two inequalities are used as constraint families in Appendix B.9, and the linear program they define is compared against the full-lattice program by `results/N3_K5_lattice.py`.

B.5 THE EXACT WORST CASE (THEOREM 8)

Throughout this subsection $K \geq 2$, $\eta \geq 1$, $f(O^*) = 1$, and

$$k_1 = (K-1)\eta + 1, \quad q = \frac{(K-1)\eta}{k_1} = 1 - \frac{1}{k_1}, \quad V_j(\eta) = 1 - q^j \left(1 - \frac{K-j}{K\eta}\right).$$

Note $k_1 \geq K \geq 2$, so $q \in (0, 1)$, and $(1-q)k_1 = 1$.

The reduced linear program. Every execution of predictive greedy induces $d_t = d_{e_t}(S^t)$ for $t = 0, \dots, K-1$ and, after enumerating $O^* = \{o_1, \dots, o_K\}$,

$$g_{t,i} = \begin{cases} d_{o_i}(S^t), & o_i \notin S^t, \\ 0, & o_i \in S^t, \end{cases} \quad 0 \leq t \leq K, \quad 1 \leq i \leq K.$$

Appendix B.9 derives, one family at a time and with the case $e_t \in O^*$ treated explicitly, that this vector satisfies

$$\begin{aligned} \sum_i g_{t,i} + \sum_{s < t} d_s &\geq 1 && (\text{sum}(t)), \\ d_t - g_{t,i}/\eta &\geq 0 && (\text{pred}(t, i)), \\ g_{t,i} - g_{t+1,i} &\geq 0 && (\text{mono}(t, i)), \\ (1 - \frac{1}{\eta})g_{t+1,i} - g_{t,i} + d_t &\geq 0 && (\text{cons}(t, i)), \end{aligned} \tag{7}$$

together with $d_t \geq 0$, $g_{t,i} \geq 0$, and that $\sum_{t < K} d_t = f(S^K)$ by telescoping. Consequently the minimum of $\sum_{t < K} d_t$ over the polyhedron equation 7 is at most the worst-case ratio $\rho_K(\eta)$: the polyhedron contains every realizable run and possibly more.

Lower bound: greedy is never worse than $\min_j V_j$. Fix a segment index j with $0 \leq j \leq K-1$ and let η lie in $[K-j, K-j+1]$ (in $[K, \infty)$ when $j = 0$). Write $M = K\eta - (K-j)$ and define nonnegative numbers $\lambda_S(t)$, $\lambda_P(t)$, $\lambda_C(t)$ (one per constraint family, the same for every index i by the symmetry of equation 7 in i) as follows. For $j \geq 1$,

$$\begin{aligned} \lambda_S(0) &= \frac{q^{j-1}M}{Kk_1}, & \lambda_S(t) &= \frac{q^{j-1-t}M}{k_1^2} \quad (1 \leq t \leq j-1), \\ \lambda_S(j) &= \frac{K-j+1-\eta}{k_1}, & \lambda_C(t) &= \frac{q^{j-1-t}M}{Kk_1} \quad (t \leq j-1), \\ \lambda_C(t) &= \frac{K-1-t}{K(\eta-1)} \quad (t \geq j), & \lambda_P(t) &= \frac{\eta - (K-t)}{K(\eta-1)} \quad (t \geq j), \end{aligned} \tag{8}$$

with $\lambda_S(t) = 0$ for $t > j$ and $\lambda_P(t) = 0$ for $t < j$; for $j = 0$,

$$\begin{aligned} \lambda_S(0) &= \frac{1}{\eta}, & \lambda_S(t) &= 0 \quad (t \geq 1), \\ \lambda_C(t) &= \frac{K-1-t}{K(\eta-1)}, & \lambda_P(t) &= \frac{\eta - (K-t)}{K(\eta-1)} \quad (0 \leq t \leq K-1). \end{aligned} \tag{9}$$

All multipliers of the family mono are 0: the lower bound does not use that the optimal gains decrease along the run.

Nonnegativity. On the segment, $\eta \geq K-j \geq 1$ and $K\eta > K-j$, so $M > 0$; since also $q \in (0, 1)$ and $k_1 > 0$, the three multipliers built from powers of q , namely $\lambda_S(0)$, $\lambda_S(t)$ for $1 \leq t \leq j-1$ and $\lambda_C(t)$ for $t \leq j-1$, are positive. For $\lambda_S(j)$, the segment condition $\eta \leq K-j+1$ is exactly $K-j+1-\eta \geq 0$, and equality holds at the right endpoint of the segment. For $\lambda_C(t)$ with $t \geq j$, $K-1-t \geq 0$ because $t \leq K-1$. For $\lambda_P(t)$ with $t \geq j$, $\eta - (K-t) \geq \eta - (K-j) \geq 0$ by the segment condition, with equality at $t = j$ at the left endpoint. The denominator $\eta-1$ is positive whenever $j \leq K-2$, since then $\eta \geq K-j \geq 2$. The one remaining case is $j = K-1$, whose segment is $[1, 2]$: there the range $t \geq j$ contains only $t = K-1$, where the numerator $K-1-t$ vanishes identically and $\lambda_C(K-1)$ is defined to be 0, while $\lambda_P(K-1) = (\eta-1)/(K(\eta-1))$ is defined by cancellation to be $1/K$. So equation 8 has no singularity on any segment.

The weighted-sum identity. For an arbitrary vector $(d_t, g_{t,i})$, not necessarily feasible, consider

$$\begin{aligned} \Sigma := & \sum_{t=0}^j \lambda_S(t) \left[\sum_i g_{t,i} + \sum_{s < t} d_s - 1 \right] + \sum_{t=j}^{K-1} \sum_i \lambda_P(t) \left[d_t - \frac{g_{t,i}}{\eta} \right] \\ & + \sum_{t=0}^{K-1} \sum_i \lambda_C(t) \left[\left(1 - \frac{1}{\eta}\right) g_{t+1,i} - g_{t,i} + d_t \right]. \end{aligned} \quad (10)$$

The last summand of the third group is void, because $\lambda_C(K-1) = 0$ by its own formula in equation 8 and equation 9. Collecting coefficients in Σ gives three families of scalar identities. The coefficient of $g_{t,i}$, for $0 \leq t \leq K$, is $\lambda_S(t) - \lambda_P(t)/\eta - \lambda_C(t) + (1 - \frac{1}{\eta})\lambda_C(t-1)$, with the conventions $\lambda_C(-1) = 0$, $\lambda_C(K) = 0$ and $\lambda_S(t) = \lambda_P(t) = 0$ outside the ranges listed above; the coefficient of d_t is $\sum_{s > t} \lambda_S(s) + K(\lambda_P(t) + \lambda_C(t))$; the constant is $-\sum_{t \leq j} \lambda_S(t)$. The claim is that these equal 0, 1 and $-V_j$ respectively, so that $\Sigma = \sum_{t < K} d_t - V_j$ identically. Each is checked below; nothing is left to "direct expansion".

(a) *The coefficient of $g_{t,i}$ vanishes.* Take $j \geq 1$ first. For $t = 0$ one has $\lambda_C(-1) = 0$ and $\lambda_P(0) = 0$ (as $0 < j$), so the coefficient is $\lambda_S(0) - \lambda_C(0) = 0$, the two entries of equation 8 being literally the same number. For $1 \leq t \leq j-1$, again $\lambda_P(t) = 0$, and dividing the coefficient by the positive quantity $q^{j-1-t}M/(Kk_1)$ turns it into

$$\frac{K}{k_1} - 1 + \left(1 - \frac{1}{\eta}\right)q = \frac{K - k_1}{k_1} + \frac{\eta - 1}{\eta} \cdot \frac{(K-1)\eta}{k_1} = \frac{(K-1)(1-\eta) + (\eta-1)(K-1)}{k_1} = 0,$$

using $K - k_1 = K - (K-1)\eta - 1 = (K-1)(1-\eta)$. For $t = j$, write $u = K - j$, so that $\eta \in [u, u+1]$ and $M = K\eta - u$, and multiply the coefficient by $K\eta k_1(\eta - 1) > 0$:

$$\underbrace{K\eta(\eta-1)(u+1-\eta)}_{\text{from } \lambda_S(j)} - \underbrace{k_1(\eta-u)}_{\text{from } \lambda_P(j)/\eta} - \underbrace{(u-1)\eta k_1}_{\text{from } \lambda_C(j)} + \underbrace{(\eta-1)^2(K\eta-u)}_{\text{from } (1-1/\eta)\lambda_C(j-1)}.$$

The two middle terms combine as $-k_1[(\eta-u) + (u-1)\eta] = -k_1u(\eta-1)$, so with $A = \eta - 1$ the whole expression is $A[K\eta(u+1-\eta) + A(K\eta-u) - k_1u]$, and the bracket expands to

$$K\eta u + K\eta - K\eta^2 + K\eta^2 - K\eta - \eta u + u - k_1u = u[K\eta - \eta + 1 - k_1] = u[k_1 - k_1] = 0.$$

For $j+1 \leq t \leq K-1$ one has $\lambda_S(t) = 0$, and multiplying by $K\eta(\eta-1)$ the coefficient becomes $-(\eta-K+t) - \eta(K-1-t) + (\eta-1)(K-t) = -\eta + K - t - \eta K + \eta + \eta t + \eta K - \eta t - K + t = 0$.

For $t = K$ only the last term survives and $\lambda_C(K-1) = 0$. The multiplication by $\eta - 1$ in the last two computations is legitimate on every segment with $j \leq K-2$, where $\eta \geq 2$; on the segment $j = K-1$, which is $[1, 2]$, both sides of each identity are rational functions of η that are continuous at $\eta = 1$ after the cancellation recorded above, so the identity at $\eta = 1$ is the limit of the identity on $(1, 2]$. For $j = 0$ the first computation is replaced by the single case $t = 0$: multiplying by $K\eta(\eta-1)$, $K(\eta-1) - (\eta-K) - \eta(K-1) = K\eta - K - \eta + K - \eta K + \eta = 0$, and the cases $1 \leq t \leq K-1$ and $t = K$ are the ones just done.

(b) *The coefficient of d_t equals 1.* For $t \geq j$ the sum $\sum_{s > t} \lambda_S(s)$ is empty, and

$$K(\lambda_P(t) + \lambda_C(t)) = K \cdot \frac{(\eta - K + t) + (K - 1 - t)}{K(\eta - 1)} = \frac{\eta - 1}{\eta - 1} = 1,$$

which at $j = K-1$, $\eta = 1$ reads $K(\frac{1}{K} + 0) = 1$ with the cancelled values. For $t = j-1$ (so $j \geq 1$), $\lambda_P(j-1) = 0$ and the only surviving λ_S is $\lambda_S(j)$, so the coefficient is

$$\frac{K-j+1-\eta}{k_1} + K \cdot \frac{M}{Kk_1} = \frac{(K-j+1-\eta) + (K\eta - K + j)}{k_1} = \frac{(K-1)\eta + 1}{k_1} = 1.$$

For $t \leq j-2$ the value is unchanged from $t+1$ to t : subtracting the two coefficients and using $\lambda_P(t) = \lambda_P(t+1) = 0$,

$$\begin{aligned} \lambda_S(t+1) + K[\lambda_C(t) - \lambda_C(t+1)] &= \frac{q^{j-2-t}M}{k_1^2} + K \cdot \frac{Mq^{j-2-t}}{Kk_1} (q-1) \\ &= \frac{q^{j-2-t}M}{k_1^2} - \frac{Mq^{j-2-t}}{k_1^2} = 0, \end{aligned}$$

because $q - 1 = -1/k_1$. Downward induction from $t = j - 1$ therefore gives the value 1 for every $t \leq j - 1$ as well.

(c) *The constant equals $-V_j$.* For $j \geq 1$, the middle multipliers form a geometric sum with ratio q , which is summed with $1 - q = 1/k_1$:

$$\sum_{t=1}^{j-1} \lambda_S(t) = \frac{M}{k_1^2} \sum_{m=0}^{j-2} q^m = \frac{M}{k_1^2} \cdot \frac{1 - q^{j-1}}{1 - q} = \frac{M(1 - q^{j-1})}{k_1},$$

so that, using $M + (K - j) = K\eta$ and hence $M + (K - j + 1 - \eta) = K\eta - \eta + 1 = k_1$,

$$\sum_{t=0}^j \lambda_S(t) = \frac{1}{k_1} \left[\frac{Mq^{j-1}}{K} + M - Mq^{j-1} + (K - j + 1 - \eta) \right] = \frac{1}{k_1} \left[k_1 - Mq^{j-1} \frac{K-1}{K} \right].$$

The definition of q gives $\frac{K-1}{K} = \frac{qk_1}{K\eta}$, so $Mq^{j-1} \frac{K-1}{K} = \frac{Mq^j k_1}{K\eta}$ and

$$\sum_{t=0}^j \lambda_S(t) = 1 - \frac{Mq^j}{K\eta} = 1 - q^j \frac{K\eta - (K - j)}{K\eta} = 1 - q^j \left(1 - \frac{K - j}{K\eta} \right) = V_j.$$

For $j = 0$ the sum is the single term $\lambda_S(0) = 1/\eta$, and $V_0 = 1 - (1 - \frac{K}{K\eta}) = 1/\eta$.

Conclusion of the lower bound. By (a), (b), (c), identity equation 10 reads $\sum_{t < K} d_t - V_j = \Sigma$ for every vector $(d_t, g_{t,i})$. At a point feasible for equation 7 every bracket in Σ is nonnegative and every multiplier is nonnegative, so $\Sigma \geq 0$ and $\sum_{t < K} d_t \geq V_j$. Since every run of predictive greedy at error η gives such a feasible point with objective $f(S^K)/f(O^*)$,

$$\rho_K(\eta) \geq V_j(\eta) \quad \text{on the segment of } j.$$

Which V_j is smallest. For $0 \leq i \leq K - 1$,

$$\begin{aligned} V_i - V_{i+1} &= q^i \left[q - \frac{q(K - i - 1)}{K\eta} - 1 + \frac{K - i}{K\eta} \right] \\ &= q^i \left[-\frac{1}{k_1} + \frac{1}{K\eta} \left(1 + \frac{K - i - 1}{k_1} \right) \right] = \frac{q^i (K - i - \eta)}{K\eta k_1}, \end{aligned}$$

where the middle step uses $q - 1 = -1/k_1$ and $(K - i) - q(K - i - 1) = 1 + \frac{K - i - 1}{k_1}$, and the last step uses $k_1 - K\eta = 1 - \eta$. On the segment $\eta \in [K - j, K - j + 1]$ the sign of $K - i - \eta$ is nonnegative for $i \leq j - 1$ and nonpositive for $i \geq j$, so the sequence $V_0, \dots, V_{K-1}$ decreases up to index j and increases after it: $\min_i V_i = V_j$ there, and neighbouring segments agree at the integer endpoints. The breakpoints are therefore the integers $2, \dots, K$, and $\rho_K(\eta) \geq \min_i V_i(\eta)$ for every $\eta \geq 1$. At $\eta \geq K$ the active index is $j = 0$ and $\min_i V_i = V_0 = 1/\eta$. The closed forms displayed in Theorem 8 for $K \in \{2, 3, 4\}$ are the specializations of $\min_j V_j$; for instance at $K = 3$ and $j = 2$, $V_2 = 1 - \frac{4\eta^2}{(2\eta+1)^2} \cdot \frac{3\eta-1}{3\eta} = \frac{16\eta+3}{3(2\eta+1)^2}$.

The identities of this proof are machine-checked by `results/N1_dual_certificate.py`, and the $K \in \{2, 3, 4\}$ closed forms by `results/T3_K3_closed_form.py`.

Upper bound: the value V_j is attained. Fix $K \geq 2$, $0 \leq j \leq K - 1$, and η with $\eta \geq K - j$, and fix any split $\eta_u \eta_o = \eta$ with $\eta_u, \eta_o \geq 1$ (the scripts use $\eta_u = \eta_o = \sqrt{\eta}$). Partition a ground set of $n = 2K$ elements into

$$C \ (|C| = j), \quad P \ (|P| = K - j), \quad O \ (|O| = K),$$

write $x = |S \cap C|$, $z = |S \cap P|$, $y = |S \cap O|$, and put

$$\delta_j = \frac{q^j}{K\eta}, \quad \tilde{\delta}_j = \eta_o \delta_j = \frac{q^j}{K\eta_u}, \quad W(0) = \frac{k_1}{K\eta_u}, \quad W(y) = \frac{(K - y)\eta_o}{K} \ (1 \leq y \leq K),$$

and $\chi(y) = \mathbf{1}[y < K]$. The instance is

$$f(S) = 1 - q^x \left(1 - \frac{y}{K} \right) + z \delta_j \chi(y), \quad \tilde{f}(S) = W(0) - q^x W(y) + z \eta_o \delta_j \chi(y). \quad (11)$$

Both are counting functions of (x, z, y) , so Lemma 14 applies with $r = 3$. Two normalizations are immediate: $f(\emptyset) = 1 - 1 = 0$ and $\tilde{f}(\emptyset) = W(0) - W(0) = 0$.

Marginal gains. Directly from equation 11, for $y < K$,

$$\Delta_C f = q^x(1-q)\frac{K-y}{K}, \quad \Delta_P f = \delta_j, \quad \Delta_O f = \begin{cases} q^x/K, & y \leq K-2, \\ q^x/K - z\delta_j, & y = K-1, \end{cases}$$

and at $y = K$ all three gains are 0. On the predictor side, $\Delta_C \tilde{f} = q^x(1-q)W(y)$ and $\Delta_P \tilde{f} = \eta_o \delta_j$, while

$$\Delta_O \tilde{f} = \begin{cases} q^x(W(0) - \frac{(K-1)\eta_o}{K}), & y = 0, \\ q^x\eta_o/K, & 1 \leq y \leq K-2, \\ q^x\eta_o/K - z\eta_o\delta_j, & y = K-1. \end{cases}$$

Two identities are used repeatedly: $(1-q)k_1 = 1$ and

$$W(0) - \frac{(K-1)\eta_o}{K} = \frac{(K-1)\eta + 1}{K\eta_u} - \frac{(K-1)\eta_u\eta_o}{K\eta_u} = \frac{1}{K\eta_u}. \quad (12)$$

Monotonicity and submodularity of f . The three first differences are nonnegative: $\Delta_C f \geq 0$ and $\Delta_P f > 0$ are immediate from $q \in (0, 1)$, and at $y = K-1$

$$\Delta_O f = \frac{q^x}{K} - z\frac{q^j}{K\eta} \geq 0 \iff \eta q^{x-j} \geq z,$$

which holds because $x \leq j$ makes $q^{x-j} \geq 1$ and $z \leq K-j \leq \eta$ by the standing assumption $\eta \geq K-j$. This is the only place where the segment condition is consumed, and it is where the breakpoints come from. The nine second differences are: $\Delta_C \Delta_C f = -q^x(1-q)^2 \frac{K-y}{K} \leq 0$; $\Delta_O \Delta_C f = -q^x(1-q)/K \leq 0$ for every $y \leq K-1$ (at $y = K-1$ the gain drops from $q^x(1-q)/K$ to 0, the same decrement); $\Delta_C \Delta_O f = -q^x(1-q)/K \leq 0$; $\Delta_P \Delta_C f = \Delta_C \Delta_P f = \Delta_P \Delta_P f = 0$; $\Delta_O \Delta_P f = 0$ for $y \leq K-2$ and $-\delta_j \leq 0$ at $y = K-1$; $\Delta_P \Delta_O f = 0$ for $y \leq K-2$ and $-\delta_j \leq 0$ at $y = K-1$; $\Delta_O \Delta_O f = 0$ for $y \leq K-3$ and $-z\delta_j \leq 0$ at $y = K-2$. Lemma 14 then gives that f is monotone and submodular.

The error is exactly (η_u, η_o) . Divide each predicted gain by the corresponding true gain, using equation 12 in the case $y = 0$ of direction O . Every single-element ratio $\tilde{d}/d$ takes one of three values:

$$\frac{\Delta_C \tilde{f}}{\Delta_C f} \Big|_{y=0} = W(0) = \frac{k_1}{K\eta_u}, \quad \frac{\Delta_O \tilde{f}}{\Delta_O f} \Big|_{y=0} = \frac{1}{\eta_u}, \quad \eta_o \text{ in all remaining cases,}$$

the remaining cases being direction C with $y \geq 1$, direction P , and direction O with $y \geq 1$. The three are ordered by $k_1 - K = (K-1)(\eta-1) \geq 0$ and $K\eta - k_1 = \eta - 1 \geq 0$, that is

$$\frac{1}{\eta_u} \leq \frac{k_1}{K\eta_u} \leq \frac{K\eta}{K\eta_u} = \eta_o.$$

The outer two are attained, so $\max \tilde{d}/d = \eta_o$ and $\max d/\tilde{d} = \eta_u$: the pair sits exactly on the band of Definition 1, and its error is not merely bounded by η .

The optimum. $f(O) = f(0, 0, K) = 1$, and for every S with $|S| \leq K$ one has $f(S) \leq 1$: at $y = K$ the value is exactly 1, and at $y \leq K-1$, $f(S) = 1 - q^x \frac{K-y}{K} + z\delta_j \leq 1$ because $z\delta_j \leq (K-j) \frac{q^j}{K\eta} \leq \frac{q^x}{K} \leq q^x \frac{K-y}{K}$ whenever $y \leq K-1$, using $z \leq K-j \leq \eta$ and $q^j \leq q^x$. So $f(O^*) = 1$ with $O^* = O$.

The per-step gain table. The run is the following; the table lists, at each state, the true and predicted gains of the three kinds of candidate.

step t	state (x, z, y)	candidate	true gain	predicted gain	picked
$t < j$	$(t, 0, 0)$	$e \in C$	q^t/k_1	$q^t/(K\eta_u)$	yes
		$e \in P$	$q^j/(K\eta)$	$q^j/(K\eta_u)$	no
		$e \in O$	q^t/K	$q^t/(K\eta_u)$	no (tie)
$t \geq j$	$(j, t-j, 0)$	$e \in P$	$q^j/(K\eta)$	$q^j/(K\eta_u)$	yes
		$e \in O$	q^j/K	$q^j/(K\eta_u)$	no (tie)

The entries follow from the gain formulas above: at $y = 0$ a C -step has true gain $q^t(1 - q) = q^t/k_1$ and predicted gain $q^t(1 - q)W(0) = q^tW(0)/k_1 = q^t/(K\eta_u)$; a P -step has predicted gain $\eta_o\delta_j = q^j/(K\eta_u)$; an O -step at $y = 0$ has predicted gain $q^x/(K\eta_u)$ by equation 12.

The greedy induction. Claim: for $0 \leq t \leq K$ the state after t steps is $(x, z, y) = (\min(t, j), \max(t - j, 0), 0)$. It holds at $t = 0$. Assume it at $t < j$. By the table the maximum predicted gain at that state is $q^t/(K\eta_u)$, attained by every remaining element of C and every element of O , while every element of P scores $q^j/(K\eta_u)$, which is strictly smaller because $q \in (0, 1)$ and $j > t$. Adversarial tie breaking picks an element of C , so the state becomes $(t + 1, 0, 0)$. Now assume the claim at t with $j \leq t < K$. The state is $(j, t - j, 0)$, C is exhausted, and the predicted gains of the remaining P elements and of the O elements are both $q^j/(K\eta_u)$: the step is a tie and the adversary picks an element of P , which is available because $t - j < K - j = |P|$. After K steps $S^K = C \cup P$ and

$$f(S^K) = 1 - q^j + (K - j)\delta_j = 1 - q^j + \frac{(K - j)q^j}{K\eta} = 1 - q^j \left(1 - \frac{K - j}{K\eta}\right) = V_j(\eta),$$

which is $V_j(\eta)f(O^*)$ because $f(O^*) = 1$. Choosing j with $\eta \in [K - j, K - j + 1]$ (that is j the minimizing index of the previous paragraph) gives $\rho_K(\eta) \leq \min_i V_i(\eta)$, which together with the lower bound is Theorem 8.

The role of the ties. Every one of the K steps above is a tie with an element of O , so adversarial tie breaking is load bearing. On the steps $t \geq j$ the tie is forced by the band: the prediction constraint of equation 7 holds with equality at the optimum, so $\eta_o d_P = d_O/\eta_u$, and the largest value the band allows a P element and the smallest it allows an O element coincide. Under strict preference the same value is an infimum rather than a maximum: replacing η by $\eta' < \eta$ inside f while declaring the band at η makes all preferences strict and gives ratio $V_j(\eta') \rightarrow V_j(\eta)$ as $\eta' \uparrow \eta$.

The identities of this proof are machine-checked by `results/N2_check.py`.

B.6 THE CEILING (THEOREM 10)

Fix $\eta_u, \eta_o \geq 1$, $K \geq 1$, $n \geq 2K$, and a constant $c > 0$.

The construction. Let $\tilde{f}(S) = c|S|$, which does not depend on the hidden optimum, and for a K -set O with $B = N \setminus O$ let

$$f_O(S) = c \left(\frac{|S \cap B|}{\eta_o} + \eta_u |S \cap O| \right).$$

f_O is modular with positive coefficients, hence monotone and submodular with $f_O(\emptyset) = 0$. Its single-element gains are c/η_o on B and $c\eta_u$ on O , against the constant predicted gain c , so the ratios $\tilde{d}/d$ take the two values η_o (on B) and $1/\eta_u$ (on O), and the error is exactly (η_u, η_o) with both factors attained. Because both functions are modular, the gain of any pair (A, B') is the sum of single-element gains along any chain from A to $A \cup B'$, so every all-pairs ratio is a weighted average of the two single-element values and the all-pairs error is exactly (η_u, η_o) as well. Since $\eta_u \geq 1 \geq 1/\eta_o$, the K largest coefficients are those of O and $f_O(O^*) = f_O(O) = c\eta_u K$.

Deterministic algorithms. The transcript of a deterministic algorithm querying $\tilde{f}$ is the same for every O , so its output T , $|T| \leq K$, is fixed before O is chosen. With $n \geq 2K$ there is a K -set $O \subseteq N \setminus T$, and for it

$$\frac{f_O(T)}{f_O(O^*)} = \frac{c|T|/\eta_o}{c\eta_u K} \leq \frac{cK/\eta_o}{c\eta_u K} = \frac{1}{\eta_u \eta_o} = \frac{1}{\eta}.$$

Randomized algorithms. Let O be uniform among K -subsets, independent of the random string. The output T is again independent of O , and writing $m = |T| \leq K$, $\mathbb{E}|T \cap O| = mK/n$ and $\mathbb{E}|T \cap B| = m(1 - K/n)$, so

$$\mathbb{E} \left[\frac{f_O(T)}{f_O(O^*)} \right] = \frac{m(1 - \frac{K}{n})/\eta_o + \eta_u mK/n}{\eta_u K} = \frac{m}{K} \left[\frac{1 - \frac{K}{n}}{\eta_u \eta_o} + \frac{K}{n} \right] \leq \left(1 - \frac{K}{n}\right) \frac{1}{\eta} + \frac{K}{n},$$

and $\min_O \leq \mathbb{E}_O$ turns this into a statement about one instance.

The matching upper bound by exhaustive search. For any S , enumerate $S = \{s_1, \dots, s_m\}$ and telescope both functions along the same chain $S_0 = \emptyset \subset S_1 \subset \dots \subset S_m = S$. The upper band applied to each of the m terms gives $\tilde{f}(S) = \sum_i \tilde{d}_{s_i}(S_{i-1}) \leq \eta_o \sum_i d_{s_i}(S_{i-1}) = \eta_o f(S)$, and the lower band gives $\tilde{f}(S) \geq f(S)/\eta_u$. Let $\hat{S}$ maximize $\tilde{f}$ over all sets of size at most K . Then

$$f(\hat{S}) \geq \frac{\tilde{f}(\hat{S})}{\eta_o} \geq \frac{\tilde{f}(O^*)}{\eta_o} \geq \frac{f(O^*)}{\eta_u \eta_o} = \frac{f(O^*)}{\eta},$$

the middle inequality by the choice of $\hat{S}$. So the constant $1/\eta$ is reached by an algorithm that is unrestricted in queries but makes no use of structure, which is the sense in which the ceiling is met.

No script checks the computations of this proof.

B.7 ASYMPTOTICS (COROLLARY 11)

Fix $\eta \geq 1$. For L_K , the substitution $u = 1/(\eta K) \in (0, 1]$ and the expansion $\ln(1 - u) = -u - u^2/2 - \dots$ give $K \ln(1 - \frac{1}{\eta K}) = -\frac{1}{\eta} + O(1/K)$, so $L_K(\eta) \rightarrow 1 - e^{-1/\eta}$; moreover $(1 - \frac{x}{K})^K$ increases in K for $x \in (0, 1]$, so the convergence of $L_K(\eta)$ is monotone from above. For $\rho_K = \min_j V_j$, fix η and let K grow. The minimizing index is the unique j with $\eta \in [K - j, K - j + 1]$, that is $j = K - \lfloor \eta \rfloor$ for noninteger η and either endpoint value at an integer η , so $K - j$ stays bounded while $j \rightarrow \infty$. With $k_1 = (K - 1)\eta + 1$,

$$\ln q^j = j \ln\left(1 - \frac{1}{k_1}\right) = -\frac{j}{k_1} + O\left(\frac{j}{k_1^2}\right), \quad \frac{j}{k_1} = \frac{K - \lfloor \eta \rfloor}{(K - 1)\eta + 1} \rightarrow \frac{1}{\eta}, \quad \frac{j}{k_1^2} \rightarrow 0,$$

so $q^j \rightarrow e^{-1/\eta}$, while the second factor of V_j satisfies $1 - \frac{K-j}{K\eta} = 1 - \frac{\lfloor \eta \rfloor}{K\eta} \rightarrow 1$. Hence $\rho_K(\eta) = V_j(\eta) \rightarrow 1 - e^{-1/\eta}$, the same limit as L_K , and by Remark 9 the two curves differ by $O(1/K)$ at fixed η .

The limit identities of this proof are machine-checked, in the form used by Theorem 12, by `results/N5_asymptotics.py`.

B.8 BOUNDED-QUERY HARDNESS (THEOREM 12)

In this subsection "information-theoretic" is used with the following meaning, stated here at its first use: the bound depends only on the number and the size of the queries that the algorithm makes and uses no computational assumption, so nothing below is a statement about running time.

The family. Fix a design parameter $\theta \geq 1$ and put $a_\theta = 1 - \frac{1}{\theta K}$. For a hidden K -set O and $S \subseteq N$ write $x = |S \setminus O|$, $y = |S \cap O|$, and define

$$F_O(S) = \theta^{-1/2} \left(1 - a_\theta^x \left(1 - \frac{y}{K}\right)\right), \quad G_O(S) = \begin{cases} 1 - a_\theta^{x+y}, & y \leq \tau, \\ 1 - a_\theta^{x+\tau} \frac{K-y}{K-\tau}, & y \geq \tau, \end{cases}$$

together with the balanced-region profile $\hat{G}(s) = 1 - a_\theta^s$. The two branches of G_O agree at $y = \tau$. The prefactor $\theta^{-1/2}$ only fixes the split $\eta_u = \eta_o = \sqrt{\theta}$ and cancels from every ratio. The facts used below are: (F1) $G_O(S) = \hat{G}(|S|)$ whenever $|S \cap O| \leq \tau$, which is immediate from the first branch since then $x + y = |S|$; (F2) F_O is monotone submodular and the pair has single-element error exactly $\Phi(\theta) = \theta(1 + \delta(\theta))$; (F3) $\max_{|S| \leq K} F_O(S) = F_O(O) = \theta^{-1/2}$; (F4) $F_O(T)/F_O(O) = 1 - a_\theta^K$ for every K -set T disjoint from O . Facts (F2) to (F4) carry the status recorded in the comment above.

Step 1: concentration. For a fixed S with $|S| \leq K$, a uniformly random K -subset O of an n -element ground set and an integer $0 \leq \tau < K$, put $m = \tau + 1$. The event $\{|S \cap O| > \tau\}$ is the union, over the $\binom{|S|}{m}$ subsets $R \subseteq S$ with $|R| = m$, of the events $\{R \subseteq O\}$, and

$$\Pr[R \subseteq O] = \frac{\binom{n-m}{K-m}}{\binom{n}{K}} = \prod_{i=0}^{m-1} \frac{K-i}{n-i} \leq \left(\frac{K}{n}\right)^m,$$

because $(K - i)n \leq K(n - i)$ reduces to $iK \leq in$, valid for $K \leq n$. A union bound over $\binom{|S|}{m} \leq \binom{K}{m} \leq K^m/m!$ choices gives

$$\Pr[|S \cap O| > \tau] \leq \binom{K}{\tau+1} \left(\frac{K}{n}\right)^{\tau+1} \leq \frac{K^{2(\tau+1)}}{(\tau+1)! n^{\tau+1}}.$$

With $Q = n^c$ queries, $\tau = c+1$ and a further output set T_0 of size at most K , summing the per-query bound and adding $\Pr[T_0 \cap O \neq \emptyset] \leq \mathbb{E}|T_0 \cap O| = |T_0|K/n \leq K^2/n$ gives

$$\Pr[\exists i : |S_i \cap O| > \tau \text{ or } T_0 \cap O \neq \emptyset] \leq \frac{K^{2c+4}}{(c+2)! n^2} + \frac{K^2}{n} < \frac{1}{2},$$

where the last step uses $n \geq 4K^{c+2}$; then $n^2 \geq 16K^{2c+4}$ bounds the first term by $1/(16 \cdot (c+2)!) \leq 1/32$, and $K^2/n \leq 1/(4K^c) \leq 1/4$, so the sum is at most $9/32$.

Step 2: valuation on the balanced event. Let $\mathcal{A}$ be deterministic, make at most Q queries of size at most K , and output T with $|T| \leq K$. Run it against the canonical oracle that answers every query S with $\hat{G}(|S|)$, and let $S_1, \dots, S_Q$ and T_0 be the resulting queries and output; they are fixed sets, independent of O . Suppose $|S_i \cap O| \leq \tau$ for every i and $T_0 \cap O = \emptyset$. By induction on i : if the first $i-1$ answers of the real oracle G_O agree with the canonical ones, then determinism makes the i th query equal to S_i , and (F1) with $|S_i \cap O| \leq \tau$ makes the real answer $G_O(S_i) = \hat{G}(|S_i|)$, the canonical one. So all Q answers agree and the output is T_0 . Since $T_0 \cap O = \emptyset$, the counts of T_0 are $y = 0$ and $x = |T_0| \leq K$, hence by (F3) and (F4)

$$\frac{F_O(T_0)}{F_O(O^*)} = 1 - a_\theta^{|T_0|} \leq 1 - a_\theta^K = 1 - \left(1 - \frac{1}{\theta K}\right)^K = L_K(\theta),$$

using $a_\theta \in (0, 1)$, which needs $\theta K > 1$.

Step 3: which band constraint binds. With $\eta_u = \eta_o = \sqrt{\theta} s$ for an inflation factor $s \geq 1$, the band of Definition 1 holds on a given edge exactly when s is at least both of the two ratios $\Delta G/(\sqrt{\theta} \Delta F)$ and $\Delta F/(\sqrt{\theta} \Delta G)$ of that edge. Writing $a = a_\theta$, the edges with both endpoints in the balanced strip give

$$\begin{aligned} x\text{-edge, } y \leq \tau : \quad \Delta F &= \theta^{-1/2} a^x (1-a) \frac{K-y}{K}, \quad \Delta G = a^{x+y} (1-a), \quad \frac{\Delta G}{\sqrt{\theta} \Delta F} = a^y \frac{K}{K-y}, \\ \frac{\Delta F}{\sqrt{\theta} \Delta G} &= \frac{K-y}{\theta K a^y} \leq 1, \quad y\text{-edge, } y+1 \leq \tau : \quad \Delta F = \frac{\theta^{-1/2} a^x}{K}, \quad \Delta G = \frac{a^{x+y}}{\theta K}, \\ \frac{\Delta G}{\sqrt{\theta} \Delta F} &= \frac{a^y}{\theta} \leq 1, \quad \frac{\Delta F}{\sqrt{\theta} \Delta G} = a^{-y}. \end{aligned}$$

The two bounds marked ≤ 1 use $\theta \geq 1$: the second is immediate, and the first follows from $a^y \geq 1 - \frac{y}{\theta K}$, which gives $\theta K a^y \geq \theta K - y \geq K - y$. The first candidate $a^y K/(K-y)$ increases in y , since the ratio of consecutive terms is $a \frac{K-y}{K-y-1} \geq 1$ exactly when $a \geq 1 - \frac{1}{K-y}$, which holds because $a = 1 - \frac{1}{\theta K}$ and $\theta K \geq K \geq K - y$; its maximum over the strip is at $y = \tau$. The second candidate a^{-y} is maximized at $y = \tau - 1$. Hence the smallest admissible s is $s = \max\{a_\theta^\tau \frac{K}{K-\tau}, a_\theta^{1-\tau}\}$, that is $1 + \delta(\theta) = s^2$ and $\Phi(\theta) = \theta(1 + \delta(\theta))$. That no other pair of sets binds harder is the part supplied by the linear program rather than by hand.

Step 4: the design parameter, and assembly. Φ is continuous and strictly increasing on $\theta \geq 1$ when $K \geq 2\tau$ and $\tau \geq 1$: the maximum of two increasing functions is increasing, and $\theta a_\theta^{2\tau}$ has derivative $a^{2\tau} + 2\tau a^{2\tau-1}/(\theta K) > 0$ while $\theta a_\theta^{2-2\tau}$ has derivative, with $u = \theta K$, $\frac{1}{K} \left(\frac{u}{u-1}\right)^{2\tau-2} [1 - \frac{2\tau-2}{u-1}]$, positive because $u \geq K \geq 2\tau > 2\tau - 1$. Also $\Phi(\theta) \geq \theta$, so $\hat{\eta} = \Phi^{-1}(\eta) \leq \eta$ whenever hypothesis (H), $\Phi(1) \leq \eta$, holds. Hypothesis (H) is implied by $K \geq \tau(1 + \frac{2}{\ln \eta})$: with $a = a_1 = 1 - 1/K$ and $\ln a < 0$,

$$\ln \Phi(1) \leq 2 \max\left\{\tau \ln a + \ln \frac{K}{K-\tau}, (1-\tau) \ln a\right\} \leq \frac{2\tau}{K-\tau} \leq \ln \eta,$$

using $-\ln(1-t) \leq t/(1-t)$ for the first branch and $(\tau-1) \ln \frac{K}{K-1} \leq \frac{\tau-1}{K-1} \leq \frac{\tau}{K-\tau}$ for the second. Now build the family with $\theta = \hat{\eta}$, so that its single-element error is exactly η and it is admissible at that error level. Let $S_1, \dots, S_Q, T_0$ be the canonical transcript of Step 2. By Step 1 the bad event has probability below $1/2$, so at least one K -set O makes every canonical query balanced and misses T_0 ; fix such an O . Step 2 then gives $f(T) = F_O(T_0) \leq L_K(\hat{\eta})f(O^*)$ for that one instance. The random choice of O is used only to establish existence.

Step 5: randomized algorithms, by averaging. For each fixing r of the random string, the algorithm $\mathcal{R}_r$ is deterministic with canonical output $T_0^{(r)}$; let E_r be the event that all its canonical queries are balanced. Pointwise in O , the variant of Step 2 that drops the requirement $T_0 \cap O = \emptyset$ (the transcript argument is unchanged, only the value bound weakens) together with $F_O \leq F_O(O^*)$ gives

$$\frac{f_O(T)}{f_O(O^*)} \leq L_K(\hat{\eta}) + \frac{|T_0^{(r)} \cap O|}{K} + \mathbf{1}[E_r^c].$$

Averaging over a uniform O bounds the middle term by K/n and, by Step 1, the last by $K^{2c+4}/((c+2)!n^2)$; averaging over r and exchanging the two finite expectations, then using $\min_O \leq \mathbb{E}_O$, gives the additive ε_n of Theorem 12. Only the averaging direction of the minimax principle is used; no minimax theorem is invoked.

Step 6: what happens as K grows. Both branches of δ tend to 0, with $K\delta(\eta) \rightarrow \max\{2\tau(1 - \frac{1}{\eta}), \frac{2(\tau-1)}{\eta}\}$, so $\hat{\eta} \rightarrow \eta$ and $L_K(\hat{\eta}) \rightarrow 1 - e^{-1/\eta}$. At finite K the constant $L_K(\hat{\eta})$ is larger than $L_K(\eta)$, since $\hat{\eta} \leq \eta$ and L_K is decreasing, so the theorem and Proposition 4 are consistent and leave a gap of order c/K between them, which closes only in the limit.

The identities of this proof are machine-checked by `results/N5_asymptotics.py` and `results/N5_delta_at_eta_hat.py`.

B.9 VALIDITY OF THE REDUCED CONSTRAINTS

This subsection supplies the step consumed by Appendix B.5: every constraint of equation 7 is a valid inequality, so the linear program is a relaxation of the set of realizable runs and its optimum is a lower bound on the worst-case ratio for every K .

Let f be monotone submodular with $f(\emptyset) = 0$, let $\tilde{f}$ be an arbitrary set function with $\tilde{f}(\emptyset) = 0$ obeying the band of Definition 1, and let $e_0, \dots, e_{K-1}$ and $S^0 = \emptyset, \dots, S^K$ be the picks and states of a run of predictive greedy, so that $e_t \in \arg \max_{e \notin S^t} \tilde{d}_e(S^t)$. Fix an optimal set $O^* = \{o_1, \dots, o_K\}$ with $f(O^*) = 1$ (if $|O^*| < K$, set $g_{t,i} = 0$ for the unused indices; every case (a) below covers them). Put $d_t = d_{e_t}(S^t)$ and

$$g_{t,i} = \begin{cases} d_{o_i}(S^t), & o_i \notin S^t, \\ 0, & o_i \in S^t, \end{cases}$$

the second line being the *zeroing convention*, which is the device that handles a step at which greedy selects an element of O^* : from that step on the corresponding variable is 0 rather than undefined. It is consistent with reading $g_{t,i}$ as a marginal gain, since an element already in S^t contributes nothing further. Finally $r_t = 1 - \sum_{s < t} d_s$, which telescopes to $r_t = f(O^*) - f(S^t)$.

Throughout, $t \in \{0, \dots, K-1\}$, $i \in \{1, \dots, K\}$, and the three cases are (a) $o_i \in S^t$; (b) $o_i \notin S^t$ and $o_i \neq e_t$; (c) $o_i = e_t$, which is the case $e_t \in O^*$.

Lemma 15 (signs). $d_t \geq 0$ and $g_{t,i} \geq 0$ for all t, i .

Proof. Each of these numbers is a marginal gain of f or, under the zeroing convention, equal to 0. This step uses monotonicity of f . $\square$

Lemma 16 (coverage, sum(t)). $\sum_{i=1}^K g_{t,i} \geq r_t$ for every t .

Proof. Let $I_t = \{i : o_i \notin S^t\}$, so that $\sum_i g_{t,i} = \sum_{i \in I_t} d_{o_i}(S^t)$ by the zeroing convention; this is exactly where a step with $e_s \in O^*$ for some $s < t$ is absorbed, the indices of the already picked

optimal elements dropping out of the sum. Order $I_t = \{i_1, \dots, i_m\}$ and telescope

$$f(S^t \cup O^*) - f(S^t) = \sum_{\ell=1}^m \left[f(S^t \cup \{o_{i_1}, \dots, o_{i_\ell}\}) - f(S^t \cup \{o_{i_1}, \dots, o_{i_{\ell-1}}\}) \right] \leq \sum_{\ell=1}^m d_{o_{i_\ell}}(S^t),$$

one application of submodularity per term. Monotonicity gives $f(S^t \cup O^*) \geq f(O^*)$, so $\sum_i g_{t,i} \geq f(O^*) - f(S^t) = r_t$. $\square$

Lemma 17 (greedy choice inside the band, $\text{pred}(t, i)$). $d_t \geq g_{t,i}/\eta$ for every t, i .

Proof. Case (a): $g_{t,i} = 0$ and $d_t \geq 0$ by Lemma 15. Case (b): chain

$$d_t = d_{e_t}(S^t) \geq \frac{\tilde{d}_{e_t}(S^t)}{\eta_o} \geq \frac{\tilde{d}_{o_i}(S^t)}{\eta_o} \geq \frac{d_{o_i}(S^t)}{\eta_u \eta_o} = \frac{g_{t,i}}{\eta},$$

the first inequality by the upper band at (S^t, e_t) , the second by the greedy choice ($o_i \notin S^t$ makes o_i eligible at step t), the third by the lower band at (S^t, o_i) . Case (c): here $g_{t,i} = d_{o_i}(S^t) = d_{e_t}(S^t) = d_t$ and the claim reads $d_t \geq d_t/\eta$, which holds because $d_t \geq 0$ and $\eta \geq 1$. No band inequality is needed in this case, so the constraint does not degrade when greedy happens to pick an optimal element. $\square$

Lemma 18 (single-element submodularity, $\text{mono}(t, i)$). $g_{t+1,i} \leq g_{t,i}$ for every t, i .

Proof. Case (a): $o_i \in S^t \subseteq S^{t+1}$, so both sides are 0. Case (b): $o_i \notin S^{t+1}$ as well, so $g_{t+1,i} = d_{o_i}(S^{t+1}) \leq d_{o_i}(S^t) = g_{t,i}$ by submodularity in its diminishing-returns form. Case (c): $o_i = e_t \in S^{t+1}$, so $g_{t+1,i} = 0$ while $g_{t,i} = d_t \geq 0$ by monotonicity. This is the only place among the four families where the case $e_t \in O^*$ replaces a submodularity argument by a monotonicity argument. $\square$

Lemma 19 (coherence, $\text{cons}(t, i)$). $(1 - \frac{1}{\eta})g_{t+1,i} \geq g_{t,i} - d_t$ for every t, i .

Proof. Case (a): the left side is 0 and the right side is $-d_t \leq 0$. Case (b): apply Lemma 7 with $S = S^t$, $e = e_t$ and $e' = o_i$; its hypothesis $\tilde{d}_{e_t}(S^t) \geq \tilde{d}_{o_i}(S^t)$ holds because o_i is eligible at step t , and $e \neq e'$ holds by the case assumption. Part (ii) of that lemma reads $(1 - \frac{1}{\eta})d_{o_i}(S^t \cup \{e_t\}) \geq d_{o_i}(S^t) - d_{e_t}(S^t)$, which is the claim, since $S^{t+1} = S^t \cup \{e_t\}$ and $o_i \notin S^{t+1}$ give $d_{o_i}(S^{t+1}) = g_{t+1,i}$. Case (c): $g_{t+1,i} = 0$ by the zeroing convention and $g_{t,i} - d_t = d_{e_t}(S^t) - d_{e_t}(S^t) = 0$, so the constraint holds with equality. Lemma 7, which requires $e \neq e'$, is not invoked in this case, which is what closes the gap this subsection exists to close. $\square$

Lemma 20 (relaxation). For every instance $(f, \tilde{f})$ satisfying Definition 1 with $\eta = \eta_u \eta_o$, every optimal set O^* and every run of predictive greedy, the vector $(d_t, g_{t,i})$ is feasible for equation 7 and its objective value equals $f(S^K)/f(O^*)$. Hence the optimum of equation 7 is a lower bound on the worst-case ratio of predictive greedy at budget K and error η .

Proof. Feasibility is Lemmas 15 to 19. The objective is $\sum_{t < K} d_t = f(S^K)$ by telescoping, and $f(O^*) = 1$ by the normalization. Every realizable run gives a feasible point, so the minimum over the feasible set is at most the minimum over realizable runs. $\square$

Remark 21 (what is and is not claimed). Lemma 20 gives one direction only. The reverse direction, that every feasible point of equation 7 is realized by some instance, is not claimed here; the attaining instances of Appendix B.5 supply what the theorem needs instead, and the agreement of equation 7 with the full-lattice program has been checked for $K \leq 5$ and at nineteen (K, η) test points.

Remark 22 (which error measure each family tolerates). The families $\text{sum}(t)$ and $\text{mono}(t, i)$ use no error notion at all. The family $\text{pred}(t, i)$ uses the band only at the greedy state S^t , so it stays valid with η replaced by η^{tr} , and, read directly as a ratio of true gains at greedy states, with η replaced by η^{sel} . The family $\text{cons}(t, i)$ invokes Lemma 7, whose proof uses the band at the state $S^t \cup \{o_i\}$, which is off the greedy trajectory; so cons is not covered by either restricted measure without a further assumption. This is the reason the L_K bound, whose proof uses only sum and pred , is stated for η^{sel} in Proposition 4, while the exact value of Theorem 8 is stated for the global η .

Remark 23 (which hypothesis each family consumes). $\text{sum}(t)$ uses submodularity and monotonicity of f and no property of $\tilde{f}$. $\text{pred}(t, i)$ uses the greedy choice and both bands, and no property of f beyond monotonicity. $\text{mono}(t, i)$ uses submodularity, and monotonicity in case (c). $\text{cons}(t, i)$ uses the greedy choice and both bands through Lemma 7, and monotonicity in cases (a) and (c). No family uses submodularity of $\tilde{f}$, which is consistent with the predictor being an arbitrary set function.

No script checks the derivations of this subsection; the linear program they justify is compared against the full-lattice program by `results/N3_K5_lattice.py`.

B.10 INSTANCES USED IN THE EXPERIMENTS

The worst-case instances run in Section 5 are exactly the families of Appendices B.3 and B.5: the family equation 1 at $\hat{a} = 2$, and the family equation 11 for $K \in \{3, 5, 8\}$ with one η per segment j . The experiment pipeline runs them through the same lazy-greedy code as the real objectives, with the element order implementing the adversarial tie direction (B before O , and C then P before O), and reproduces the values $L_K(\hat{a})$ and $V_j(\eta)$ of the two previous subsections to 10^{-10} ; on every V_j instance it measures $\eta^{\text{sel}} = \eta$, and on every member of equation 1 it measures $\eta^{\text{sel}} = \eta^{\text{tr}} = \hat{a} < \eta$.

The computations of this subsection are machine-checked by `results/E4_worst_instances.py` and `results/F2_etaset_tight.py`.

C EXPERIMENT DETAILS AND ADDITIONAL FIGURES

C.1 PROTOCOL

Every run is a single trajectory of predictive greedy on $\tilde{f}$ of length equal to the largest budget of its family, from which all shorter budgets are read off as prefixes. Alongside it the same code runs greedy on f to provide the ratio denominator. Both objectives are wrapped in a cache keyed by the set, so a repeated query is free and the running times quoted in the released notes count distinct sets. At every state of the $\tilde{f}$ trajectory the true best gain $\max_{e \notin S^t} d_e(S^t)$ is computed as well, which is what makes η^{sel} measurable; where f is not submodular this maximum is taken by an exact scan rather than by a lazy heap, since lazy evaluation is only valid under submodularity. Steps whose chosen true gain is nonpositive are excluded from η^{sel} and counted separately, and that count is the last column of Table 1.

The trajectory error η^{tr} is measured on the same runs, but only over candidate pairs whose true and predicted gains both exceed one quantization unit of the objective, which is one over the size of the held-out split for feature selection and one covered node for influence maximization. Pairs below that threshold are exactly the ones that send a multiplicative ratio to infinity for reasons that have nothing to do with the selection, so the figures reported for η^{tr} are lower estimates of it, and the bound $L_K(\eta^{\text{tr}})$ computed from them is an upper estimate of the bound the untrimmed measure would give. Both directions favour η^{tr} , which is why the comparison with η^{sel} in Section 5 is stated in the direction it is.

For feature selection the predictor is the 5-fold cross-validation accuracy on the training split only; the held-out split that defines f is never touched by $\tilde{f}$. For influence maximization each observed graph is drawn once per (network, p , seed) and reused along the whole trajectory, so $\tilde{f}$ is a fixed function and not a fresh sample per query. For summarization the reference summaries are held out of every surrogate.

C.2 HOW GOOD IS THE OPT PROXY?

Ratios in Table 1 divide by greedy on the true objective rather than by the optimum, which makes them upper estimates. On the smallest feature-selection dataset the optimum was enumerated exactly for budgets up to $K = 5$ over 10 splits. At that budget the median of $f(S^f)/f(O^*)$ is 0.982, so using the proxy inflates a reported ratio by 1.8 % in the median; the median of $f(S^{\tilde{f}})/f(O^*)$, that is the quantity the theory bounds, is 0.942. Exhaustive enumeration was not attempted at larger budgets

or on the larger datasets, so no ratio against a true optimum is claimed anywhere else. On the largest dataset a conservative under-estimate $\hat{O}$ of the optimum was computed as the best true value over 2200 candidate subsets per split (the surrogate’s top-ranked subsets together with uniformly random ones), over 10 splits: the pool improved on greedy on f in 3 splits, never by more than 0.33 %. Since $f(\hat{O}) \leq f(O^*)$, this bounds how much better the sampled pool is, not the gap to the true optimum, and the table keeps greedy on f as the denominator.

C.3 BASELINE COMPARISON

The baseline panel uses 10 splits rather than the 30 of the main sweep, to bound running time. On the largest feature-selection dataset the median held-out accuracy of greedy on $\tilde{f}$ is at least that of the best of 4 classical selectors at 7 of the 7 budgets, all of them scored by the same f so that the comparison is between selected subsets and not between downstream models. On the smallest dataset the outcome is mixed: greedy on $\tilde{f}$ leads at some budgets and trails at others, in every case by an amount on the order of a single accuracy quantization unit of that dataset, so the fair statement there is a draw rather than an improvement.

C.4 ADDITIONAL FIGURES

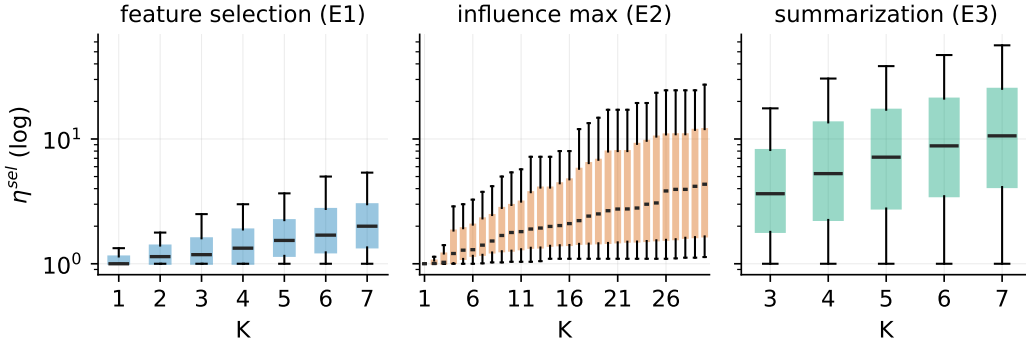


Figure 2: Distribution of the measured selection error η^{sel} by budget K for each surrogate family (boxes: quartiles over runs; log scale; outliers suppressed). The learned surrogates stay in the single digits over the whole budget range, while the heuristic surrogates do not.

C.5 CONSTRUCTED INSTANCES

The 19 constructed instances of Section 5 consist of 16 instances of the V_j family, one per segment of Theorem 8 at $K \in \{3, 5, 8\}$, and 3 instances of the per- K family of Theorem 6. On the V_j instances the measured selection error reproduces the design error to 5.3×10^{-15} ; on the family of Theorem 6 the measured η^{sel} reproduces the parameter $\hat{a}$ of that family to 1.8×10^{-15} , which is the equality $\eta^{\text{sel}} = \hat{a}$ asserted there and not the global error of those instances, which is larger. The representative row of Table 1 is the $j = 2$ instance at $K = 5$ and error 3.5, whose realized ratio is 0.278. No constructed instance produced a sign disagreement between predicted and true gains (largest measured share 0 %).

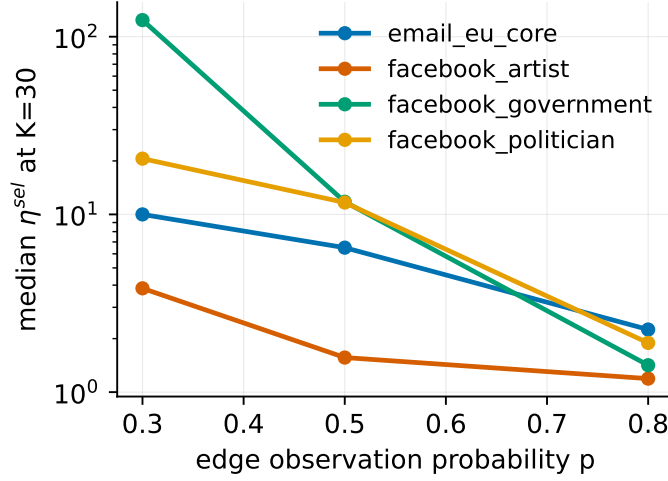


Figure 3: Median η^{sel} at $K = 30$ against the edge observation probability p , one line per network (log scale). The error rises as the graph is observed less, on all 4 networks; over the same range the realized ratio falls only from no worse than 0.988 to between 0.882 and 0.963, while the certified bound $L_{30}(\eta^{\text{sel}})$ falls by up to a factor 64.

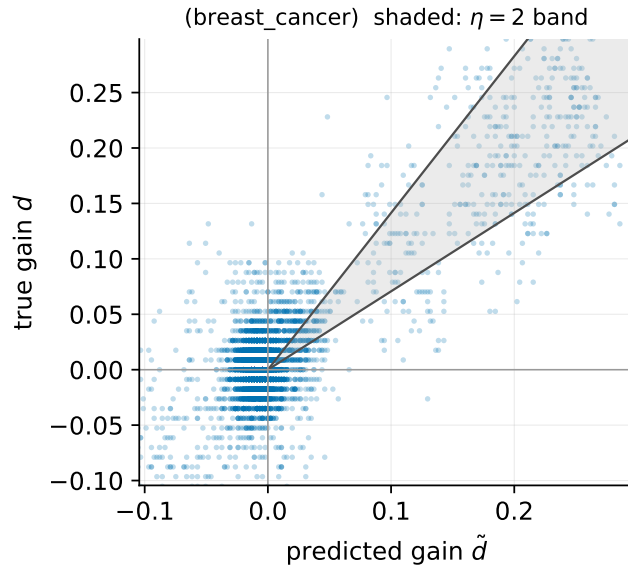


Figure 4: True gain d against predicted gain $\tilde{d}$ over the candidate evaluations of one feature-selection dataset, with a multiplicative error band shaded. Sign disagreements concentrate near zero gain, where the accuracy quantization of the objective dominates, and the large gains are ordered correctly. This is why the guarantee is read at η^{sel} , which charges only the step that was taken, rather than at the band error of every candidate pair.